

SALISHAN

BY

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RESEARCH AND THEORY
WHOSE WORDS THESE ARE, AND HOW WRONG THE CORPUS COULD BE

ORIOR
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*For Megan, my loving wife, the only person that really understands
what I mean when I speak.*

*For Autumn and Opal, without you this thought would have never
occurred.*

To everyone else, I wish you well.

ABSTRACT

The corpus everything in this work is measured against is Salishan speech, written down. This research paper says whose words it holds, how wrong it could be, and what was done to find out.

Every table opens with the person who spoke, before the linguist who published and before anyone who read the paper into a file. Twenty-five tables carry 13,510 rows. Sixteen name their speakers. Nine cite a published source and name no speaker, and those say so instead of guessing. A linguist wrote the paper and a person read the paper into a table, and neither of those is whose language it is.

How wrong the corpus could be is stated as a bound, because the checks that matter report zero and zero failures in a sample is not a rate of zero. Three channels fail for different reasons: a form written that the paper does not hold, a token the paper prints that no row holds, and a token the corpus lost on the way out of a reader. Taken as independent the joint bound is 4.47×10^{-10} per line. Section 4 of the derivation says where that independence is thin, and the largest unmodelled term is that one person read every table.

The instrument found a border it was never shown. Mellesmoen and Kye label every form they cite as northern or southern Lushootseed. The boundary sits on disk as a fact a linguist published. Given the forms and never the labels, holding the concept fixed through the word web and flattening the pooled run counts to maximum entropy, the term that carries the border comes back as the stressed schwa, southern, at 45 against 6 over the same concepts, beaten by 1 of 200 random borders.

Two papers here are set in a font with no Unicode map. Their extracted text is the font's encoding and not the page. Their drafts are rebuilt rule by rule and remain drafts: a rule that guesses at labialization is named as guessing, and a fifth repair rule was specified, measured to change nothing, and withdrawn. Five more papers hold none of their own orthography. Each contributes zero, and none is recorded as a disagreement.

These tools read a language and can put one back. That is stated plainly here. Near the center of a subject a regeneration is a copy, and far from it the output is obvious nonsense. Between them is a narrow band where the output is coherent and may already not be the language, and nothing here marks which side of it a result fell on. That question belongs to a native speaker, and no amount of measurement turns it into one an algorithm can answer.

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WHOSE WORDS THESE ARE

Purpose: Find whose language is in this corpus, and which table holds it. **Scope:** every `.oracle.tsv` under `build/oracles`

These languages belong to the people who speak them. None of this work exists without them. Everything else in this research is measured against the tables below, and the tables are their words, written down.

Each entry opens with who spoke, because that is whose language it is. A linguist wrote the paper and a person read the paper into a table, and neither of those is whose language it is. Where a paper cites a published dictionary and never says who spoke, the entry says so, and the linguist does not go in the who column.

Conditions the speakers set are recorded with them below and hold wherever this corpus is used.

Written by `maint/data/salishan/hand_extraction/pure_corpus_index.py` from `maint/data/salishan/corpus_script_extraction/paper_config.py`, the only place a speaker's name is typed. Nothing in this chapter is typed by hand, and an edit made here is lost the next time that script runs.

1.0.1 *Lushootseed*

- **Martha Lamont, Northern dialect**
- **Annie Jack, Southern dialect**

Mellesmoen_Kye_ICSNL61, 467 rows, `Mellesmoen_Kye_ICSNL61.oracle.tsv`

Both recorded by Leon Metcalf in the 1950s. The only paper here that labels every form by dialect, and the border test is scored against it.

1.0.2 *Lushootseed*

- **Susie Sampson Peter, Upper Skagit**
- **Martha LaMont, Tulalip-Skagit**

1983_Hilbert, 60 rows, `1983_Hilbert.oracle.tsv`

Vi taq^wšəblu Hilbert wrote the paper; the twenty-one examples were said by her aunt Susie Sampson Peter and by Martha LaMont, recorded by Leon Metcalf between 1950 and 1958 and by Thom Hess in 1963. The record credited Hilbert as the speaker until the hand extraction found it.

1.0.3 *St'át'imcets*

- **Ķweswapáw (Linda Redan), Qayqáyten**
- **Sam Mitchell, in van Eijk and Williams 1981**

Matthewson_Redan_ICSNL61, 104 rows, Matthewson_Redan_ICSNL61.oracle.tsv

Ķweswapáw told the story over Zoom on 31 October 2025, three minutes twenty-eight seconds, and the audio and video are held by her. Sam Mitchell is the speaker of the earlier text the paper cites.

1.0.4 *St'át'imcets*

- **Qwa7yán'ak (Carl Alexander), Nxwísten**

AlexanderDavis_ICSNL61, 541 rows, AlexanderDavis_ICSNL61.oracle.tsv

Recorded at Nxwísten on 7 July 2025, just over half an hour.

1.0.5 *Nuxalk*

- **Dr. Margaret Siwallace**

22-Nater-Bella-Coola-tale-10, 285 rows, 22-Nater-Bella-Coola-tale-10.oracle.tsv

Recorded about 1975, published 2015.

1.0.6 *n̓l̓eʔkepmxcín*

- **wlwl̓melst (Maurice Michell), Southern yutémkt dialect**

ICSNL59_LaFontaine_Janzen_final, 226 rows, ICSNL59_LaFontaine_Janzen_final.oracle.tsv

He shares his four stories freely for people connecting with the language. They came from his mother nxwelinek and his grandmother ʔústko.

1.0.7 *n̓l̓eʔkepmxcín*

- **K^wəłtəzétk^wu (Bernice Garcia), čəlétk^wu (Coldwater)**

ICSNL59_Garcia_Hannon_Stacey_final, 371 rows, ICSNL59_Garcia_Hannon_Stacey_final.oracle.tsv

She asks it be acknowledged she is a Kamloops Indian Residential School speaker re-learning her language.

1.0.8 *Mainland Comox (ayajuthem)*

- **Mary George, Sliammon**
- **Noel George Harry**
- **Tommy Paul**

ICSNL56_DavisJ_2_final-1, 1661 rows, ICSNL56_DavisJ_2_final-1.oracle.tsv

Recorded 1969 to 1980. The oracle check applies this paper's own repair and the coverage check applies none, which is one of the two drifts named above.

1.0.9 *n̄eʔkepmxcín*

- **Bev Phillips, Lytton First Nation (ł̓q̓əmcín)**

HallPhillipsICSNL60, 406 rows, HallPhillipsICSNL60.oracle.tsv

Her own reading of the story is in build/audio. It is an oracle for the extraction and not only a source. The coverage check applies a cruder space closer here than the oracle check does, the other drift named above.

1.0.10 *Nsyilxcən*

- **George Lezard, Penticton Indian Reserve**
- **Nellie Guitterez, Upper Nicola Indian Band**
- **Kiláwna? (Andrew McGinnis), Penticton Indian Reserve**

19-Lyon_ICSNL50_final-78, 1417 rows, 19-Lyon_ICSNL50_final-78.oracle.tsv

George Lezard recorded 1966 by Randy Bouchard, transcribed by Larry Pierre 1970, updated by permission of Arnie Baptiste, his son. Nellie Guitterez recorded 1978 or 1979 by Yvonne Hébert, reprinted by permission of Lynne Jorgesen, her great-granddaughter.

1.0.11 *Nsyilxcən*

- **Lottie Lindley, Upper Nicola**

2013_Lindley_Lyon, 653 rows, 2013_Lindley_Lyon.oracle.tsv

1.0.12 *Lushootseed*

- The paper names no speaker. Its forms are cited from a published source.

1975_Hilbert_Hess, 74 rows, 1975_Hilbert_Hess.oracle.tsv

A 1975 typescript, scanned, and what is on disk is OCR of the scan. The paper names no speaker for its examples. The OCR carries none of the orthography: zero schwas, zero raised w, zero barred l, zero wedges, against 169, 77, 41 and 33 in the hand extraction. It writes taqWsablu for taq^wšəblu and slahal for sləhal. All 200 direction-one disagreements filed against this paper come from that, and they are the source text and not the reader. It needs a drafted page text and a place in NOT_FAITHFUL, the way the two Lyon papers have, before the check against it means anything.

1.0.13 *n̄le?kepmxcín and Secwepemtsín*

- **Charley Alexis Mayoos**
- **William Celestin**

2012_Robertson, 195 rows, 2012_Robertson.oracle.tsv

Their texts are written in Chinuk pipa. Texts 3 to 6 are Chinook Jargon, which is a pidgin and is not Salish, and the who column keeps those out.

1.0.14 *eighteen Central Salish languages*

- The paper names no speaker. Its forms are cited from a published source.

WolfeICSNL60, 772 rows, WolfeICSNL60.oracle.tsv

Every form is cited from a published dictionary of one of eighteen languages. There is nobody this corpus is of. The who column carries the language instead, and the reader writes a .pure.tsv keyed by it and no flat pure file.

1.0.15 *Nuxalk*

- The paper names no speaker. Its forms are cited from a published source.

ICSNL59_Nater_2_final, 671 rows, ICSNL59_Nater_2_final.oracle.tsv

Nater's own records, from his 1990 dictionary and 1984 grammar. No speaker is named. Six entries and two tables are Heiltsuk, Oowekyala, Kwakwaka and Haisla, which are North Wakashan and not Salish at all.

1.0.16 *Nsyilxcən*

- **ḵmxnalqs (Delphine Derrickson-Armstrong), stq̄aʔtk^wḵniwt**
- **čəskʔáknaʔ (Dave Michele), stq̄aʔtk^wḵniwt**

LyonICSNL60_Inch-2, 537 rows, LyonICSNL60_Inch-2.oracle.tsv

Elicited from both speakers. Most cells of its two tables are starred, which is a form the linguist built and the speakers rejected, and those are held out.

1.0.17 *Twana*

- The paper names no speaker. Its forms are cited from a published source.

Kim_TwanaReduplication_final, 358 rows, Kim_TwanaReduplication_final.oracle.tsv

Every Twana form is Drachman's, out of a 1969 dissertation the paper calls the only reliable reference in existence for this. No speaker is named. Footnote 9's four Moses-Columbian forms are the only place its extraction repeats a combining mark, and those are corrected from the page.

1.0.18 *Nuxalk*

- The paper names no speaker. Its forms are cited from a published source.

2013_Nater, 2605 rows, 2013_Nater.oracle.tsv

1407 numbered entries out of Nater's own 1990 dictionary. No speaker is named.

1.0.19 *n̄leʔkepmxcín*

- **Bev Phillips**
- **ćúʔsinek (Marty Aspinall)**
- **k^wałtèzetk^w (Bernice Garcia)**

Hall-et-al_-ICSNL_61-1, 494 rows, Hall-et-al_-ICSNL_61-1.oracle.tsv

The forms are cited from Thompson and Thompson's grammar and dictionary. These three are the speakers the paper thanks, two examples are Bev Phillips reading her own story, and k^wałtèzetk^w introduces herself in the acknowledgement.

1.0.20 *St'át'imcets*

- **Qwa7yán'ak (Carl Alexander), Nxwísten**

ICSNL58_Davis_Mellesmoen_final, 441 rows, ICSNL58_Davis_Mellesmoen_final.oracle.tsv

Its data has three sources: van Eijk's dictionary, Davis et al. in preparation, and elicitation with Carl Alexander. It labels forms (U) and (L) for Upper and Lower St'át'imcets, the second external dialect label in the archive.

1.0.21 *Tillamook*

- The paper names no speaker. Its forms are cited from a published source.

1967_Hamp, 154 rows, 1967_Hamp.oracle.tsv

Eric P. Hamp, Another Look at Tillamook Phonology, ICSNL 2. It cites no words. Its language content is four consonant inventories set side by side and two feature matrices. No speaker is

named, and the paper works throughout from Thompson and Thompson, Reichard, Kinkade, Drachman and Edel. What it does record is that a Tillamook speaker was living in 1967 and it does not say who: the urgency of the matter is put as being that there is yet a surviving speaker available for possible re-check, and page 2 grants one sense of its claim only on the assumption that a last remaining speaker is typical of a community. Both sentences are in the table. The Twana chart is used, in the page's own words, without his permission, meaning Drachman's. The dot under the uvular fricatives prints solid in some cells of these charts and as an open ring in others, and the two positions swap between the Tillamook and Twana charts. It is one mark and the variation is the typewriter.

1.0.22 *n̄leʔkepmxcín*

- **Bev Phillips, Lytton (ʔqəmxcín) dialect**

ICSNL58_Givens_Hall_final, 178 rows, ICSNL58_Givens_Hall_final.oracle.tsv

Katherine Givens and Brent Hall, *The Moon and the Birchbark Canoe* (ʔ máʔxetn pe ʔ qʷʔinéwʔ), ICSNL 58. Seven pages, read one at a time off page renders. Page 1: the story was recounted in *N̄leʔkepmxcín* by Bev Phillips, a native speaker of the Lytton dialect, who also helped with the translation, and Givens and Hall transcribed and glossed it. She is quoted on the page choosing the story and saying creation stories are not just stories to us, and the paper's first footnote thanks her for entrusting it to them. The recording is held as `speech/icsnl_proceedings/ICSNL58_GivensHall_MoonAndBirch`, the only one of the three whose name carries the transcribers and not the speaker, and a reader working from that filename alone read it as a second speaker. Footnote 4 defines (VG), a volunteered gloss, as a translated sentence BP offered. The parenthesis at the right margin is the only thing separating her English from the authors', and examples 12, 13 and 14 lack it. Its section 4 repeats every sentence of section 2 in morpheme-broken form, which gave a second independent reading of every word and turned up three places where the two tiers disagree. Footnotes 2, 6, 7 and 8 each declare a mark or a parsing the authors reached by ear or by inference, which is more than any other paper here states about its own readings.

1.0.23 *Snohomish*

- The paper names no speaker. Its forms are cited from a published source.

1967_Hess, 248 rows, 1967_Hess.oracle.tsv

Thom Hess, *The Morph /-(ə)b/ in Snohomish*, ICSNL 2. Oracle.tsv checked against the paper. No speaker named. The forms are Hess's Snohomish data. Marked characters, page 3, by eye and inside one table: *xáyəb* 'laugh' and *xʷúyub* 'sell' print a bare *x* where *ʔáyəb* 'salty' prints a barred *x* body with a glottalization mark, *ʔádəb* 'bitter' prints the same *t* as the English word taste on the first line of that page with a glottalization mark added, and *ǵáyəb* 'cloud' carries a hook at the top of its *d* and a bar across the foot, one composite, where the *d* of *ʔádəb* one column away is bare. Below 16x either half of that composite drops out. Glyphs in the printed text unclear. 1975_Hilbert_Hess sets the same character in *dəxʷgʷəʔəlad*s, and Nater's Coast Salish **ʔaʔ* 'bitter, salt' matches *ʔáyəb* segment for segment.

1.0.24 *Twana*

- The paper names no speaker. Its forms are cited from a published source.

1967_Elmendorf, 276 rows, 1967_Elmendorf.oracle.tsv

William W. Elmendorf, Word Tabu and Change Rates, ICSNL 2. Thirteen languages are cited in it and Twana is its subject. That is the language named here; every row carries its own language in the who column. No speaker is named anywhere: the forms come from Boas and Haeberlin 1927, Krueger 1967, McIlwraith 1948, Walters 1938 and Ray 1932, from Warren Snyder's Suquamish list and Wayne Suttles' Squamish field notes, and from Elmendorf's own field data. This typescript writes its uvular as x under a dot where 1975_Hilbert_Hess writes it under a caron, and 1967_Hess agrees with it, which is two papers of one conference against one of a later year. One mark is unresolved and marked so in the table: a short raised stroke over the s of the Columbia ska'u on page 7, which is not the wedge the Upper Chehalis sč'a'u carries two words earlier.

1.0.25 *Columbian*

- The paper names no speaker. Its forms are cited from a published source.

1967_Kinkade, 316 rows, 1967_Kinkade.oracle.tsv

M. Dale Kinkade, Deictics in Columbian: A Work Paper, ICSNL 2. Twelve pages, read one at a time off page renders. It is about Columbian; Kalispel comes from Vogt 1940, Coeur d'Alene from Reichard 1938, and three Colville forms from an unnamed speaker who also knew Columbian. Every row carries its own language in the who column. No speaker is named anywhere in the paper: it says my Cm informants and says no more than that. The forms in this table were said by people it does not identify. Its typewriter has three raised marks and the table's symbol notes turn on telling them apart, a comma with a thick head and a curling tail, a V wedge, and a straight acute. čėñ on page 10 carries all three in one word and is the control. The wedge appears only in the Kalispel and Coeur d'Alene forms, ten times for ten, and never in the Columbian ones, whose seven caron readings are assigned and are all ǰ or the č of číłx. Whether that split is about the language or about page order is not decidable here, because the wedge first appears on page 8 and every Columbian form was typed on pages 1 to 7. 1967_Elmendorf, from the same conference, distinguishes a wedge from a short raised stroke in its own table. The distinction is one these typescripts can carry.

25 tables, 13510 rows read by hand.

THE CORPUS UNDER THE INSTRUMENT

2.1 ORIOR ON SALISHAN

Purpose: Apply the orior method to the Salishan papers: find the language, and decide what may join a pure corpus. **Scope:** data/salishan/

The method is in ../orior-method.md. This chapter records what happens when it is fed these papers.

2.1.1 *The idea*

English is the target. We want what English does not account for.

That inverts the problem into the half that is resourced. Nobody has to know Nsyilxcən or Lushootseed to find it in a paper. English has to be known, and the residue is the language, a gloss, or a page of font damage.

2.1.2 *The flow*

- | | |
|--------------------------|--|
| 1 pure target corpus | one per language, from the hand-read papers |
| 2 pure reference corpus | English, from the translations those readers marked |
| 3 sift noise | screen a paper against the reference corpus |
| 4 truthy against residue | measure the noise, and let section 3 say if it may be read |

2.1.3 *What runs it*

Directory	Holds
orior/	the algorithm: squash, total variation, split-half, support and entropy
corpus_script_extraction/	the twelve hand-written readers and their repairs and checks
orior_algorithmic_extraction/	the sift applied to the papers with no reader, and boundary_check.py
word_web/	the forms of every hand extraction joined by concept, shape and context

orior.py has nothing to tune and no per-language term. Everything else decides what to hand it.

2.1.4 Results

Thirteen papers read by hand. Twelve have a reader written against them, and every token of those twelve is accounted for:

13 of 13 papers have a hand extraction
 12 of 12 papers have every token of the language accounted for
 2314 lines of known-pure target language, across six languages

The thirteenth, 1975_Hilbert_Hess, is read and has no reader yet. `reader_check` holds it ungraded instead of counting it. The six languages are the ones with a pure corpus on disk; two of the thirteen papers are in languages that have no anchor.

Seven anchors: six languages and English. 20 of the 21 pairs are readable, at distances 0.510 to 0.835 against split-half floors of 0.157 to 0.641. The pair that does not read is Lushootseed against Nsyilxcən, 0.620 apart with the Lushootseed anchor's own floor at 0.641.

That floor is not the resolution of the estimator at 219 lines, and treating it as one was an error. `self_distance` cuts a corpus at its midpoint, and a corpus is its papers concatenated in file order. The cut lands on a paper boundary and reports the distance between whichever papers it separated. Lushootseed is two papers in two dialects and two orthographies, and the midpoint puts one on each side. Shuffling the corpus before the cut puts a mixture on both sides and the floor falls to 0.305. Every corpus moves the same way, by 1.35 to 3.64 times, and the unshuffled number is the larger every time. A resolution cannot rise when its sample grows, and this one does. That rise is the tell.

Read at the shuffled floors, all 15 pairs among the six languages separate, Lushootseed against Nsyilxcən included, and six of the 15 change verdict. The 21 above counts the same six languages plus English, which adds six more pairs.

The papers are a separate question and the anchor floors do not decide it. `language_check.py` prints the worst anchor `D_self` and then judges each paper against that paper's own split-half instead, the right thing to judge it against. Of the 106 papers naming a language the anchors know, 3 are readable and the distributions agree with the prose on all 3. Of the 103 that are not, 8 hold too small a sample to ask and 95 fail for one reason: the gap between the nearest anchor and the runner-up has a median of 0.0247, against a median paper floor of 0.3527. The gap is fourteen times inside the noise.

Sifted from the papers with no reader:

144 papers written to build/corpora/sifted
 13190 lines nearer the language, 20361 residue for a person to look at
 114 of the 144 carry the language their own front matter names

Admission on each corpus's own growth curve:

Language	Pure	Candidates	Admitted	D_self before	after	Support after
nleʔkepmxcín	451	339	339	0.3098	0.3196	1249
St'át'imcets	371	868	280	0.2768	0.3459	1520
Nsyilxcən	749	4898	1000	0.1571	0.1893	1322
Lushootseed	219	1262	1262	0.6411	0.4163	2377
Comox	426	1275	0	0.1956	0.1956	407
Nuxalk	98	1359	0	0.2536	0.2536	346

nleʔkepmxcín took every candidate and stayed where it was. Comox and Nuxalk refused all of theirs: tipping the whole set into Comox takes it from 407 cells to 2586 and its split-half distance from 0.196 to 0.457, which is a second distribution arriving instead of more of the first.

Lushootseed took all 1262 and its split-half distance fell from 0.641 to 0.416. The admission rule is only a ceiling, and a corpus sitting above the band it belongs in will admit whatever pulls it down. What was pulling it down here is now known: 0.641 is two dialects on opposite sides of the midpoint cut, and 1262 candidates dropped into the middle of them dilute that split without teaching the anchor anything. A falling floor was read as a corpus improving when it was a boundary being blurred.

Seven languages hold candidates and have no hand-read corpus to grow: Halkomelem 739, Montana Salish 136, Twana 72, Secwepemctsin 33, Straits 23, Squamish 17, Upper Chehalis 14.

2.1.5 *What it cannot do*

Support is still climbing on every one of these corpora. None has seen its own alphabet. A candidate that looks nothing like a member is not thereby wrong, and resembling what is already there is the wrong test.

The sift finds candidates. It does not read them. The corpus is still hand extracted and verified in three passes, for line and for notation, and this exists to say where to look.

Growing the Lushootseed anchor does not reach the 103 papers, and an earlier version of this document said it would. The 95 that are large enough to ask fail because a paper's residue sits almost the same distance from all six language anchors: the median gap between the nearest and the runner-up is 0.0247 and the median paper floor is 0.3527. Nothing about the anchors' own sizes moves those two numbers past each other.

What the 95 need is a measure that separates the anchors as seen from a mixed residue, which the byte pair distribution does not. `corpus-derivation.md` section 7 has the first thing that does: holding a concept fixed with the word web's gloss edge, flattening the pooled counts to maximum entropy, and testing one run at a time. That recovers a published dialect border inside Lushootseed at a corpus size where the whole-distribution comparison needs 2.9 times more text. It has not yet been turned on the 95.

Two dialects pooled into one anchor is still worth splitting, because it inflated the floor from 0.305 to 0.641 and it is the same defect in every corpus that holds more than one paper. Splitting it is a correction to the anchors, not a route to the 95.

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HOW WRONG IT COULD BE

Purpose: Say how wrong the pure corpus could be, from what the checks have actually seen, and say what the number rests on. **Scope:** `maint/data/salishan/corpus_derivation.py`, which writes this file, and the three checks it reads.

Rewritten by python `maint/data/salishan/corpus_derivation.py` after every change to a hand extraction. Nothing in it is typed by hand.

3.0.1 1. *What is being estimated*

For one line of the pure corpus, the probability that it is not what it claims to be: not the target language, or not what the paper printed. Every number below is an upper bound on that, never a measurement of it, because the checks that matter report zero and zero failures in a sample is not a rate of zero.

With no failures in N independent trials the true rate is under $3/N$ with 95 percent confidence. That bound is the rule of three, and it turns a clean check into a number. Where a check did see a failure, the observed rate is reported instead, because it is larger than any bound would be.

3.0.2 2. *What the checks have seen*

A paper contributes trials only where its extracted text matches what the page prints. The others are checked against a damaged source. Their disagreements measure the source and not the table. The references chapter names each of them.

Why the left panel is here. The bound in Section 3 is a single number and a single number cannot show whether it came from one clean paper or from 17. The panel puts every paper on the picture so the shape of the evidence is visible: how much each one contributed, and which ones failed anything. A reader should come away able to say which papers the number rests on.

What to read off it. Every paper is one point, trials against failures. The 16 on the zero line are the evidence. The 9 above it are checked against a text that is not what their page prints. The reader squares are a separate measurement in their own color, and they sit high: a reader is a script and it gets a great deal wrong. Section 5 gives those per paper, the only form they mean anything in.

Why the right panel is here. The extraction has a lifetime, and the question is which reader carries it. A person reads a paper at a fixed accuracy however large the corpus gets. That arm is flat. The algorithm's accuracy is a function of corpus size. That arm climbs. Whether and where

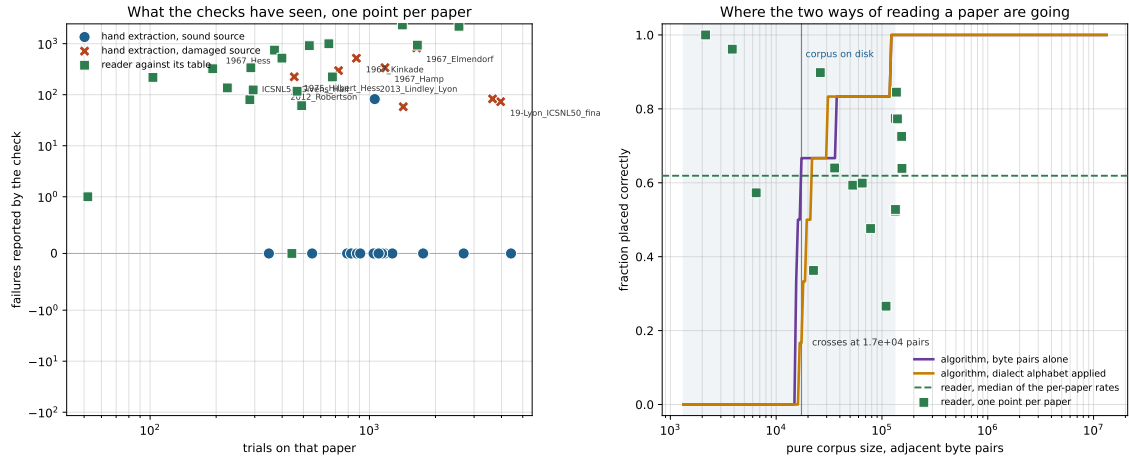


Figure 3.1: Hand extraction against its paper

they cross decides whether the corpus is worth growing for its own sake, and the panel answers it.

What to read off it. The algorithm arm is the fraction of dialects whose distance from the other corpora pooled into one reference corpus clears twice their resolution. The arm starts at zero, because a small corpus resolves nothing, and it crosses the reader arm at $1.73\text{e}+04$ byte pairs, which is behind the corpus already on disk. Both arms are drawn: byte pairs alone, and byte pairs with the dialect’s own alphabet applied. Setting an algorithm given nothing beside a person given a page is not a fair comparison, and the alphabet is the first part of what the extraction already knows that it can be handed.

Paper	Rows	Forms checked	Not in paper	Tokens checked	No row holds	Counts
Mellesmoen_Kye_	467	437	0	354	0	yes
ICSNL61						
1983_Hilbert	60	445	0	103	0	yes
Matthewson_Redan_	104	225	0	123	0	yes
ICSNL61						
AlexanderDavis_	541	1476	0	282	0	yes
ICSNL61						
22-Water-Bella-	285	579	0	313	0	yes
Coola-tale-10						
ICSNL59_LaFontaine_	226	829	0	349	0	yes
Janzen_final						
ICSNL59_Garcia_	371	784	0	488	0	yes
Hannon_Stacey_final						
ICSNL56_DavisJ_2_	1661	2131	0	557	0	yes
final-1						
HallPhillipsICSNL60	406	1118	0	639	0	yes
19-Lyon_ICSNL50_	1417	2788	41	1175	32	no
final-78						
2013_Lindley_Lyon	653	2424	46	1218	37	no
1975_Hilbert_Hess	74	429	200	25	24	no
2012_Robertson	195	1324	38	102	20	no
WolfeICSNL60	772	763	0	378	0	yes
ICSNL59_Water_2_	671	762	0	282	0	yes
final						
LyonICSNL60_Inch-2	537	666	0	435	0	yes
Kim_Twana	358	528	0	297	0	yes
Reduplication_final						
2013_Water	2605	2524	0	1896	0	yes
Hall-et-al_-ICSNL_	494	508	0	369	0	yes
61-1						
ICSNL58_Davis_	441	483	0	425	0	yes
Mellesmoen_final						
1967_Hamp	154	1176	337	1	1	no
ICSNL58_Givens_Hall_	178	896	80	161	2	yes
final						
1967_Hess	248	707	281	16	16	no

1967_Elmendorf	276	1445	631	197	196	no
1967_Kinkade	316	871	518	1	1	no

17 of the 25 hand extractions are checked against a sound source. Together they put 15154 distinct written forms and 7451 language tokens through the two directions.

3.0.3 3. The channels

A wrong line arrives through one of three channels. They are separate because they fail for different reasons. The first is a person writing a form the paper does not hold, the second is a person walking past a form the paper does hold, and the third is the corpus losing a token on its way out of a reader.

The reader counts are not a channel here. A reader is written for one paper, and what it gets wrong is a fact about that paper and not a draw from a rate the next paper shares. Pooling 18 of them into one denominator would report a rate that nothing is sampling. They are in Section 5 per paper.

Channel	Failures	Trials	Bound
a form written that the paper does not hold	80	15154	0.00528
a token in the paper that no row holds	2	7451	0.000268
a token the corpus lost on the way out	3	9512	0.000315

A line has to pass all three. Taking them as independent, the joint bound is the product:
4.47e-10 per line

How much of that number has settled

Three significant figures is only a format. The digit worth reporting has stopped moving, and the way to find it is to watch the bound as each paper joined the corpus.

Papers counted	Joint bound
1	1.835e-08
2	7.042e-09
3	4.421e-09
4	1.275e-09
5	7.64e-10
6	4.667e-10
7	2.955e-10
8	1.6e-10
9	1.103e-10
10	9.008e-11
11	7.685e-11
12	6.458e-11
13	5.744e-11
14	3.294e-11
15	3.002e-11
16	2.731e-11
17	4.469e-10

No digit has settled. The bound falls with every paper because every paper adds trials and none has yet added a failure. Quoting 4.47e-10 as though the 85 meant something would be reporting the format. The table shows the extraction still buying accuracy at the rate of about one order of magnitude every 10.5 papers, and the digit to report will settle when that stops.

Where 1e-26 lands

The target this file was asked for is 1e-26 per line over the whole extraction. It is not reached and it is not close, and the honest form of the answer is the distance.

Channel	Trials now	Trials at 993 papers	Bound then
a form written that the paper does not hold	15154	885171	3.39e-06
a token in the paper that no row holds	7451	435226	6.89e-06
a token the corpus lost on the way out	9512	555612	5.4e-06

Reading all 993 papers of the archive, and finding nothing wrong in any of them, takes the joint bound from $4.47e-10$ to about $1.26e-16$. That is 10 orders of magnitude short of $1e-26$.

Closing the rest by counting is not available. Each channel would have to reach about $1e+09$ trials, which is roughly $2e+03$ times the whole archive. There is no reading schedule that gets there, and a file claiming $1e-26$ from these three channels would be reporting a number nothing measured.

What a number that small would actually need is more channels that fail independently, not more trials in these three. Section 7 is one: a term recovered from the forms and scored against a border a linguist published, which fails for a reason none of the three share. Independent channels multiply, and that is how an exponent like this one is reached. Section 4 is where the independence is doubted, and it should be read before this number is quoted anywhere.

3.0.4 4. What the number rests on

The independence is the weak part and it is weak in three named places.

- **One person read every table.** The three channels catch different kinds of mistake but they do not catch a systematic misreading of one orthography, because the same reading produced the row and the expectation. This is the largest unmodelled term and no amount of trials touches it.
- **Direction one and direction two share a source.** Both ask questions of the same extracted text. A paper whose text is wrong in a way nobody has noticed fails both at once, and for that reason the papers with a known-damaged source are excluded from the count instead of given a worse bound.
- **Every channel runs through one codebase.** `salish_marking.py` and `salish_unsorted.py` decide what counts as a language token, and all three channels ask them. A defect in either is common to all three at once, and two such defects have already been found this way. Both are in the references chapter.

What would move the number honestly is a second person reading a table that has already been read. That is the addition that fails for a reason none of the three share, and until it exists the first bullet stands above every number in this file.

3.0.5 5. Readers against their tables

Paper	Rows asked for	Reproduced	Items written	Invented	Wrong language
Mellesmoen_Kye_ICSNL61	443	443	1638	0	0
1983_Hilbert	52	50	323	1	0
Matthewson_Redan_ICSNL61	103	59	343	217	1

AlexanderDavis_ICSNL61	532	193	1484	914	0
22-Nater-Bella-Coola-tale-10	285	256	337	80	0
ICSNL59_LaFontaine_Janzen_final	225	144	347	136	0
ICSNL59_Garcia_Hannon_Stacey_final	369	219	1198	748	0
ICSNL56_DavisJ_2_final-1	1654	991	1625	574	362
HallPhillipsICSNL60	399	190	961	521	0
19-Lyon_ICSNL50_final-78	1413	376	7171	2134	193
2013_Lindley_Lyon	653	341	4551	996	0
2012_Robertson	193	101	435	323	0
WolfeICSNL60	678	525	909	223	0
ICSNL59_Nater_2_final	288	152	533	337	0
LyonICSNL60_Inch-2	491	415	603	61	0
Kim_TwanaReduplication_final	295	228	423	124	0
2013_Nater	2560	1857	4039	2189	0
Hall-et-al_-ICSNL_61-1	468	299	777	117	0

The readers get a great deal wrong. The median reproduces 0.619 of what its table asks for, and the spread runs from one paper to the next with no common rate behind it, because each reader was written to one paper's layout. These are a table in Section 3 instead of a term for that reason.

A reader that does not reproduce a row is not by itself an impurity. The row is in the hand extraction either way, and the extraction is the oracle. The last two columns count what the reader added, the part that can reach the pure stream without a person having written it.

3.0.6 6. The word web

`maint/data/salishan/word_web/word_web.py` joins every form in the hand extractions to the forms it is related to, and writes one file per group under `build/corpora`. The web has three kinds of edge, each measured off the extraction and none of them listed by hand.

- **concept**, two forms whose glosses share a content word. This is the edge that crosses an orthography, because the gloss is the only part of a form that two papers wrote the same way.
- **shape**, two forms of one group sharing a leading or trailing run of four characters. Salish morphology is heavily affixed and reduplicating. A shared run is usually a shared root or affix, and the shape edge measures that run without parsing it.
- **context**, two forms written in the same section of the same paper by the same speaker.

The web makes an anchor a concept expressed as a distribution instead of a bag of characters. The byte pair distribution cannot see that two orthographies wrote one word, and the concept edge is where that is recorded.

3.0.7 7. The dialect border

Lushootseed is not one dialect. The northern and southern varieties have known land and family borders, and Mellesmoen and Kye's stress paper labels every form it cites with which one it came from. The hand extraction copied that into the `who` column. The border sits on disk as a fact published by a linguist.

That makes it something a test of this algorithm almost never has: an answer that did not come from the algorithm. `maint/data/salishan/orior_algorithmic_extraction/boundary_check.py` loads the labels, sets them aside, and only compares at the end.

Comparing whole distributions does not work at this size. The method resolves at 6707 bytes and the two varieties hold 1469 and 2768. Neither the byte pair distribution nor the word web separates them, and a blind partition scores no better than the majority class. The check prints what that route would need: about 2.9 times the labeled Lushootseed now on disk.

Asking for one term at a time does work, and it is the same move the sound work uses. Because both varieties are one language, a distribution over their runs measures almost only what they have in common, and at this size that shared mass swamps the difference. Flattening the pooled counts to maximum entropy takes it out: a run is weighed against where the pooled total alone would put it. A run carrying no border information then contributes nothing however common it is. Each run is its own test, and what a run needs is enough of itself, not enough of the language.

The concept is held fixed while this is asked, over the 58 concepts both varieties name, using the word web's gloss edge. Without that control a run can separate the two sets because the varieties differ or because different words were cited, and those are not the same finding.

Run width	Runs tested	Terms found	Expected by chance	Random borders that matched it
2	75	2	0.20	1 of 200
3	51	0	0.14	200 of 200
4	23	0	0.06	200 of 200

Term	Deviate	Northern	Southern
ə	-3.76	6	45
əʌ	3.18	6	0

The last column of the first table is the test that is allowed to fail. The border is put back on the same forms at random 200 times and the radix run again on each, and at width 2 only 1 of those random borders found as much as the published one. Swapping the two sides also negates every deviate exactly, but this estimator does that on any two sets whatever and it is evidence of nothing.

The term carrying the border is the stressed schwa, and it is southern: 45 of them to 6 in the north over the same concepts. The paper those labels came from is Mellesmoen and Kye's comparative analysis of stress in northern and southern Lushootseed. The algorithm was shown the forms and never the labels, and it returned the paper's subject.

Compiled By: dstroyo (Douglas Quigg) <dquigg123@gmail.com> **Generated by:** `maint/data/salishan/corpus_derivation.py`

REPRODUCING WHAT THE LINGUISTS CLAIMED

4.1 NATER ON BELLA COOLA

Purpose: Take a claim a linguist published about one of these languages, and find out whether the corpus on disk reproduces it. **Scope:** `build/experiments/nater_bella_coola.py`, the Nuxalk corpora under `build/corpora/`.

Margaret Siwallace told the tale this is measured over. Hank Nater wrote it down and published it, and a person read it into a table for this work. The language is hers.

4.1.1 *Why this is a different kind of test*

Almost every reading in this work is scored against the corpus it came out of. Section 7 of corpus-derivation is the exception that mattered most: a border a linguist published, held out, and recovered. This is the second of that kind and it is cheaper, because the answer was printed in a table forty years ago and nothing here can move it.

Nater’s ICSNL59 paper, *Voiceless Words in Bella Coola: Fact vs. Fiction*, disputes Mellesmoen (2021) on four points. Its Table 1 prints the Bella Coola phoneme inventory with a per centum distribution across the entire morpheme inventory, from Nater (1984:27). Thirty-four phonemes with a percentage each. Transcribed off the page they sum to 99.98. That sum is how the transcription is checked: a table with a row dropped still looks like a table, and one that sums to 100 does not have a row missing.

That table is an oracle in the strict sense this tree uses. It is a measurement somebody else made, over a sample this work has never seen, and no reading taken here can adjust it.

4.1.2 *What had to be thrown out first*

Three Nuxalk corpora sit in `build/corpora/`. Two of them cannot carry this measurement and neither says so in its name.

One is nearly a tenth English. The purity control is a word list built from `cc_english.txt`, four letters and up, which is already in the tree for the sift’s own anchor. How Salish is Bella Coola scores 169 English tokens in 1934, at 0.087: *introduction, however, cognates, database*. The other two score 0.018 and 0.008.

The mark set is not the test used here, and using it would have been the obvious mistake. A mark set calls a token the language when it carries one of the characters the language is written

with, and Ayajuthem and St'át'imcets are written in practical orthographies that use none of them. Asked that way, Mary George's corpus reads 0.639 foreign and Carl Alexander's 0.602, when both are the language. An English word list has the opposite failure and it is the harmless one: it can only accuse a token of being English, and a Bella Coola word is not in an English dictionary.

One is Nater's own cluster charts. The pure stream for the voiceless-words paper holds ck, cp, cq, kx' i: those are the entries of his Tables 2 to 5, lifted out of the paper by its reader. Counting phonemes over that file counts an obstruent cluster inventory as though it were speech, and pooling it with the narrative inflated every obstruent while leaving a five per cent short.

It is told from speech by measuring it. The tale runs 8.50 phonemes to a word with 0.954 of its words carrying a vowel; the chart runs under three phonemes with under a third of its entries vowelised. A chart has that shape and a language does not. The check is in the script and not in a list of filenames. The next chart to arrive is caught by the same test.

Margaret Siwallace's tale remains: 217 words, 644 phonemes read as a morpheme inventory, 31 characters no phoneme accounts for.

4.1.3 *The reproduction*

Counted as a morpheme inventory, the way Nater counted it, the tale sits at a total variation of 0.169 from his table.

Three of his rows read exactly zero in the corpus: the vocalic sonorants $\mathfrak{m}$, $\mathfrak{n}$ and $\mathfrak{l}$. The corpus does not write the syllabicity diacritic and defines $\mathfrak{m}$ as $\mathfrak{m}$. That is a fact about the extraction and not about Bella Coola, and it puts 3.58 per cent of the published table in cells the corpus can never fill while inflating the three plain sonorants by the same mass.

Folding his syllabic rows into their plain counterparts is a correction to the oracle. Doing it without saying so would make this worthless. Made openly, it predicts where the three plain sonorants should land, and they land there:

Phoneme	Nater, folded	The tale	Difference
$\mathfrak{m}$	4.99	5.59	+0.60
$\mathfrak{n}$	4.91	5.43	+0.52
$\mathfrak{l}$	4.82	4.50	-0.32

The total variation falls from 0.169 to **0.136**, and that fall of 0.033 measures what the three unwritten diacritics were costing.

Several rows agree closely without any correction at all. $\mathfrak{a}$, the most frequent phoneme in the language, reads 15.06 against 15.84. $\mathfrak{i}$ reads 7.92 against 8.12. $\mathfrak{x}'$ reads 0.93 against 0.95, $\mathfrak{x}^w$ 1.09 against 1.17, $\mathfrak{c}'$ 1.86 against 1.78.

4.1.4 *Whether 0.136 is anything*

A total variation is a number until it has a floor under it and a wrong answer beside it.

The floor. The tale is split by alternating words, which samples it instead of cutting it in place, and the two halves are measured against each other. That is the resolution of this count at 644 phonemes: **0.066**. The reading sits at about twice its own floor. There is a real gap between this corpus and the published table and it is not all noise.

The wrong answer. The same count is taken over every other Salishan corpus on disk and scored against Nater's Bella Coola table. Fourteen of them are large enough to ask.

Corpus	Phonemes	From Nater's table
Margaret Siwallace, Bella Coola	644	0.136
Delphine Derrickson Armstrong and Dave Michele	1001	0.239
Mary George	3295	0.299
George Lezard and others	3709	0.299
Martha Lamont and Annie Jack	550	0.315
Lottie Lindley	4138	0.318
Susie Sampson Peter and Martha LaMont	728	0.360
Carl Alexander	3106	0.392
wlwlme1st and Maurice Michell	1130	0.392
Charley Alexis Mayoos and William Celestin	373	0.397
Linda Redan	700	0.407
Kweltezetkwu and Bernice Garcia	1333	0.414
Bev Phillips	1530	0.444

The nearest language that is not Bella Coola sits at 0.239, and several of those corpora are five times the size of the tale. The gap from Bella Coola to the runner-up is 0.103, which is a floor and a half. A 217-word tale, read out of a PDF by a script and checked by hand, lands nearer a phoneme table published in 1984 than fourteen other Salishan corpora do.

That is a positive control on the corpus and not on the method. What it says is that the extraction, the reader and the three checks preserved enough of Bella Coola for a frequency count to find its way home.

4.1.5 *The claim about nuclei*

Mellesmoen's Good Enough Nucleus Hypothesis, as Nater quotes it, holds that every word has at least one syllable and that anything equal or greater in sonority to a fricative may stand in the nuclear position. Nater disputes the exclusion of stops, cites all-stop words and roots, and concludes with Newman (1947) that Bella Coola is largely non-syllabic.

Whether such words are in this corpus is a count. In Margaret Siwallace's tale, **17 distinct morphemes carry no sonorant and no vowel**: ck, cx, c'kt, kł, k^wc, k^wχ', k'x, sk^w', sx, tx, tχ, tχ^w, t'x^w, xł, x^wc among them. One of the 17 is built from stops alone: k^wk^w'.

The cluster chart answers this question 103 times over and the answer is worth nothing, because every entry in it is an obstruent cluster by construction. Reciting the table a claim was argued from is not evidence for the claim. The narrative is the only line here that bears on it, and it rests on one short tale.

This counts occurrences and argues nothing about syllable theory, which nothing here reaches. What can be said is that voiceless words are in the running speech on disk, at 17 in a single tale. They are not an artifact of citation lists.

4.1.6 *What this cannot do*

The areal claim is out of reach. Nater's fourth point is that long obstruent clustering is shared with Oowekyala, at Rivers Inlet, and is therefore areal and not inherited. Oowekyala is Wakashan and there is no Oowekyala on disk. The test has a shape: the same phoneme count over an Oowekyala corpus, against Bella Coola and against the Salishan corpora, with the question being whether Bella Coola sits nearer its neighbor than its family. Nothing here can run it until that corpus exists, and saying so is as far as this goes.

The residual is not explained. After the fold, s is 3.32 low, ʔ 2.01 high, u 2.36 high and k^w 1.90 high. s is the most frequent obstruent in the language and one 217-word tale is a thin sample of it.

Whether that gap is the sample, the genre, or a difference between Nater's morpheme inventory and any running narrative is not decided here, and more Bella Coola narrative would decide it.

A morpheme inventory built from marked boundaries is not the one Nater had. His was a lexicographer's list. This one holds what the corpus marks with a hyphen and a shelf, deduplicated. They are the same kind of object and they are not the same object.

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FROM THE PAGE TO A DISTRIBUTION OVER SOUNDS

Purpose: Follow one Salishan word from the mark on the page to a symbol a distribution can be taken over, and say which stage each tool in this tree belongs to. A reader measuring anything on this corpus then knows what their symbols stand for. **Scope:** `maint/data/salishan/corpus_script_extraction/salish_marking.py`, `maint/data/salishan/corpus_script_extraction/paper_config.py`, `build/experiments/salish_phonemes.py`, the hand extractions under `oracles/` in the corpus.

This chapter exists because the question keeps being asked and answered in conversation. The answer lives in a header comment in `build/experiments/`, which is not where anybody looks, and the reader who needs it is the reader who does not yet know it is there.

5.1 THREE STAGES, AND THEY ARE NOT INTERCHANGEABLE

A symbol in this corpus can mean three different things, and a distribution taken over each of them answers a different question.

1. A **character** is a codepoint in a file. A distribution over characters is a distribution over what a transcriber typed.
2. A **segment** is a base with everything that modifies it. A distribution over segments is a distribution over the units the orthography writes.
3. A **sound** is the referent of the segment, with the definitions of one sound collapsed. A distribution over sounds is the first one that is about the language.

Each stage discards something the stage before it carried, and each discard is a claim that has to be true. Skipping a stage does not leave the measurement neutral. It leaves it measuring the thing that stage was there to remove.

5.2 STAGE ZERO, THE PAGE

Nothing above starts until somebody has read the paper. A text extraction of a scanned type-script is not the paper: all four ICSNL 2 papers held here carry none of their own orthography in their text layer, zero of the nine marks the corpus uses, and `ORTHOGRAPHY_ABSENT` in `maint/data/salishan/hand_extraction/papers.py` names them so that no check grades a correct reading against a file the reading is right to differ from.

The trap runs the other way as well. *The Moon and the Birchbark Canoe* is born digital and its text layer is faithful, but the embedded font draws U+026C, the l with a belt, as an l with a bar through the stem. A reader working from page renders sees l and writes l. The paper's text layer holds U+026C 112 times and U+0142 not once. The glyph on the page and the character the page is drawing are two different questions, and the hand extraction records the second with the first written down beside it.

5.3 STAGE ONE, CHARACTERS

TEXT_SPACE is the union of every character these orthographies write their languages with (salish_marking.py:88). It is built from MARKED, the marked consonants and the schwa (salish_marking.py:30), PRACTICAL, the digit 7 (salish_marking.py:76), and the stacked glottalization marks, the labialization modifier and Nater's raised space.

Two things about it are worth stating plainly, because both have caused a wrong measurement here.

It is a membership test and not an alphabet. Its job is to answer whether a token is in one of these languages. A per-paper set is written as this plus what that paper adds, never as that paper's own alphabet, and the reason is in the file: a test built on one paper's characters reads every other paper as holding no language, and then reports nothing missing from them either.

It carries no phonetic content at all. ɬ and ʔ are both in it, as separate members, because both occur. Nothing in TEXT_SPACE says they are one sound.

Typographic ligatures are repaired before anything else, without a test, because fi is fi and no decision is involved (salish_marking.py:56). That one was found by a case sensitive match turning up five against five.

5.4 STAGE TWO, SEGMENTS

^w is not a segment. It is labialization on the consonant before it. ' is not a segment either; it is glottalization on the base it sits over. Counting per character measures two halves of one segment as though they were independent.

The rule is to join every combining mark and every modifier letter to the base in front of it. The hand extractions already do this when they write a form. Applied to Bev Phillips's *When Old One Created the Earth*, it turns 5552 characters over 35 distinct into 4839 segments over 50 distinct. The inventory grows under clustering instead of shrinking, because k^w and k stop being one letter counted twice. Twenty of the fifty carry a mark or a modifier:

č k^w k k^w m n p q^w q q^w w x^w x x^w y z ž é é^w é^w

é is one of the fifty. It is the segment that carried the Lushootseed dialect border. At the character level it is two codepoints, and the radix found it at width two for that reason.

There is no shared function in this tree that performs this clustering. The rule is stated in prose, followed by hand in the oracles, and implemented once inside build/experiments/pooled_sound.py. Anyone measuring segments today writes it again.

5.5 STAGE THREE, SOUNDS

`build/experiments/salish_phonemes.py` holds the table that takes a definition and returns the sound it stands for. Every asking of this question arrives at that file, and nothing in the tree points to it.

The definitions that differ across the corpus, counted by segment over the hand extractions that existed when the table was built, which was 23 of them:

Sound	How the corpus writes it
lateral fricative	ʎ in seventeen papers, ʎ in four. LaFontaine and Janzen write ʎ 329 times and ʎ never. Nater's voiceless paper writes ʎ 98 times and ʎ three.
glottal stop	ʔ in most, the digit 7 in the van Eijk orthography. Alexander and Davis write 7 934 times and ʔ four. Matthewson and Redan write 7 125 times and ʔ twice. A capital ʔ occurs in one paper.
uvular fricative	χ, ʁ and x, all three in six or more papers.
schwa	ə at U+0259 and ə at U+01DD, a turned e and a different codepoint.

Three of those four were named in the refs chapter. The fourth came out of counting, and so did the capital glottal stop and the aspirated p^h and t^h that only Nater writes. That is the argument for building the table by counting the corpus instead of from a description of it.

5.5.1 *Where the correspondences come from*

Nater states his own, in a paper this corpus holds: $/ʎ/ = /ʎ/, /χ/ = /χ/, /χ^w/ = /χ^w/, /p', t', q'/ = /p', t', q/, /k^w', q^w'/ = /k^w, q^w/$ and $/'/ = [ʔ]$. A linguist writing down what his own symbols equal is the strongest source available here. The van Eijk 7 is declared in the tree already. The rest follow the standard descriptions of Americanist transcription, they were assigned by hand, and any error in them is an error in every result taken downstream.

5.5.2 *Two errors that table had, both caught by reading its own output*

The first draft sent k^w and q^w to one symbol. A labialized velar and a labialized uvular are different phonemes in these languages, and the collapse showed itself at once: $k^wʎ$, $qʎ$ and $q^wʎ$ arrived as one form, which merges three words.

The second draft gave the uvular fricative the symbol x . A plain velar x already passes through as that same symbol. The mapped uvular and the unmapped velar then met in one cell. A pass-through is not free of the namespace it passes into.

Every sound now takes a codepoint of its own from the private use area, assigned by the table and never typed. No sound can collide with a letter the corpus writes. Both errors are recorded instead of quietly fixed, because a table like this is exactly where a wrong assertion is invisible: nothing downstream can tell a real merger from a typo.

5.5.3 *What the mapping does not claim*

Mapping ʎ and ʎ to one symbol says two definitions are one sound. That is a fact about transcription practice. Whether Nuxalk's lateral fricative and Lushootseed's are the same sound is a question for a comparativist, and nothing here asserts it. What the mapping buys is narrow and it is enough: a difference measured afterward is not the difference between two typists.

5.6 WHAT A DISTRIBUTION READS AT EACH STAGE

Over what	What the number is about
bytes	the encoding, and whether a mark is precomposed or decomposed
characters	the transcriber, and which conference volume the paper appeared in
segments	the orthography, comparable inside one paper and across papers sharing a convention
sounds	the language, comparable across every paper in the corpus

The failure this ordering exists to prevent has a shape worth naming. A model counting character bigrams reads two dialects that share a word as maximally far apart, because it is measuring the transcriber and not the speaker. The distance is real, it is stable, it will replicate, and it is about the wrong thing.

5.7 WHAT IS NOT BUILT

Three gaps, stated so that nobody has to rediscover them.

The phoneme table lives in an experiment. It is under build/experiments/ and nothing in the engine imports it. Work needing sounds today either finds that file or writes the table again.

The clustering rule has no shared implementation. Stage two is prose plus one copy inside one experiment.

The word web is still built on written forms. The refs chapter states this and it is unchanged: nothing takes a character out of TEXT_SPACE and hands back what it stands for at the point the web is built. Two dialects writing one word two ways are still two nodes. Stage three would close it.

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THE RECORDINGS UNDER THE INSTRUMENT

6.1 WHAT THE PROSODY FIELD HOLDS

Purpose: Ask the question the workbook records as unasked, whether the prosody residual identifies the speaker, and report what the sound representation turned out to hold instead. **Scope:** `src/engine/python/representation/sound/perceived_sound.py`, `maint/analysis/sound/binary_sound.py`, `build/experiments/speaker_residual.py`, `build/experiments/rotation_cost.py`.

Note, 26 September. `perceived_sound.py` was deleted with `src/engine/python/` by commit 74601c1 on 26 September and is in no file at orior 1948ae1. `binary_sound.py` is at the pin and imports from that tree. What this chapter reports was run before the deletion. **Later, 26 September.** Commit d09b489 (orior PR 12) put back the 126 files 74601c1 deleted, each byte-identical to its state before the deletion, and biohub pins orior at d09b489. Every file named here that 74601c1 deleted is at the pin again; nothing here was rerun against them.

Bev Phillips, of Lytton First Nation, spoke all three of these recordings, in *nłe?kepmxcín*, in the Lytton (*łqəmčín*) dialect. *When Old One Created the Earth* is an original *spıləxm* of hers, written and told by her, and she is an author of that paper and of *You Will Be Sorry* with Brent Hall. She chose the third story and helped translate it, and Katherine Givens and Brent Hall transcribed and glossed it. The language is hers and are two of the three papers.

6.1.1 *The question*

The workbook states it and leaves it open. Its wording is that pitch against the speaker's own median is a per-speaker constant subtracted, the center of mass again. What remains in that field is the residual, and whether that residual identifies the speaker has not been asked.

The field is already built. `perceived_sound.py` keeps prosody apart from the segment code deliberately, because a field made shift invariant would put a stressed form and an unstressed one at one address. It holds four things per 10 ms frame: loudness, pitch against the speaker's own median, how that pitch is moving, and voicing. Each is thresholded at its own median over the recording. Each bit splits the frames in half by construction and what a distance between two recordings reads is the joint arrangement of the four and never their levels.

6.1.2 *What was made*

`binary_sound.py` wrote the representation of all three recordings.

Recording	Seconds	Frames	Prosody, from flat
ICSNL58 The Moon and the Birchbark Canoe	234.9	23489	0.3708
ICSNL59 You Will Be Sorry	832.5	83249	0.3498
ICSNL60 When Old One Created the Earth	1350.0	134997	0.4594

The 24 bit segment field reads 0.99 from flat on every one of them, and that figure is the frame count and not the speech: 16.8 million states against 134997 frames. The module's own header says and the narrow widths are what it asks to be read.

These are also the first recordings held here at the length the vocalization work said it needed. The workbook records that true unit scale was not reachable with the earlier clips, that a one minute clip gives 1053 symbols for a whale and 281 for a bird, and that measuring the arrangement of units needs tens of minutes. Two of these are 13.9 and 22.5 minutes.

6.1.3 *The answer, through the necessary condition*

Only one speaker is held. There is no discrimination question to ask. The askable question is the condition underneath it: a residual that identifies a speaker has to hold still within one. Three recordings of Bev Phillips should then sit near the floor of the estimator.

That is Proposition 1's own shape, and it is why the thin sample is enough. A necessary condition cannot confirm identification, and its failure settles the matter.

Every recording is trimmed to 23489 frames, the shortest, because a null taken at one length bounds nothing about another. The floor is the recording split by alternating frames and measured against itself, which samples it instead of cutting it in place. A midpoint cut would put the first half of a narrative against the second and report the distance between two passages, the defect the corpus chapter records inflating every anchor floor by 1.35 to 3.64 times.

		Apart
The Moon and the Birchbark Canoe	When Old One Created the Earth	0.1437
The Moon and the Birchbark Canoe	You Will Be Sorry	0.1807
You Will Be Sorry	When Old One Created the Earth	0.3112

The three floors are 0.0098, 0.0166 and 0.0077. The three distances run **9 to 19 floors**.

The residual does not hold still within one speaker. It does not identify her. One person, one language, one dialect, and her three recordings sit nine to nineteen times the resolution apart on the field built to hold what is left after her own pitch range is divided out.

6.1.4 *What that does not settle*

Three narratives in three ICSNL volumes vary together. The story, the session, the microphone and the encode all move with the recording and nothing here separates them. The measurement shows that the field does not hold still across them, and not which of those moved it.

The failure is also one directional in the useful way. Had the three sat at the floor, that would have been consistent with the residual identifying her and consistent with the field holding nothing at all, and a second speaker would have been needed to tell those apart. They do not sit at the floor, and no second speaker is needed to read that.

6.2 THE SEGMENT FIELD IS NOT COMPARABLE BETWEEN RECORDINGS

6.2.1 *What the code does*

represented rotates the segment measurement before thresholding it. decorrelated takes the right singular vectors of the centered measurement, and the measurement it is handed is one recording. So segment bit 3 of one recording and segment bit 3 of another are different axes, and a distance between two recordings' segment profiles is partly a distance between two rotations.

The module explains why it rotates, which is that neighboring bands rise and fall together and thresholding them as they come gives bits that restate each other. It does not say the basis belongs to the recording, and nothing downstream of it compares two recordings. The question had not come up.

6.2.2 *A check that was not a check*

The first attempt at measuring this compared two halves of one recording, which share a rotation, against two whole recordings, which do not. It gives ratios of 8.8 to 1.6 across the widths and looks like a confirmation.

It is not one. The prosody field is never rotated and gives 18.6 by the same measure. Two recordings differ for every other reason as well, and a check both arms pass is measuring something both arms have. It is recorded here because it read as a result while it was being written.

6.2.3 *What isolates it*

One recording, cut in half once, and the two halves compared twice. In the first arm represented runs on the whole recording and the frames are cut afterwards. One rotation computed over both halves is used for both. In the second, each half of the samples goes through represented alone and carries a rotation computed over itself. The audio is identical and every other step is identical.

The gap is the rotation.

Recording, at 6 bits	One basis	Two bases	The rotation
The Moon and the Birchbark Canoe	0.0995	0.3376	+0.2381
You Will Be Sorry	0.0769	0.2324	+0.1556
When Old One Created the Earth	0.0617	0.1430	+0.0813

At six bits the rotation multiplies the distance between two halves of one recording by 2.3 to 3.4 times. It costs +0.15 to +0.28 at eight, ten and twelve bits as well.

The measured distances between recordings' segment fields at six bits are 0.2309, 0.2346 and 0.2819, the same size as the rotation cost on its own. **A segment distance taken between two recordings is reading their bases.** The field is not comparable across recordings without a shared rotation, and every segment row in the previous section is withdrawn as a reading of speech.

Nothing else in this tree compares two recordings' segment fields. No published number moves. What changes is that the next thing to do so has to fit one rotation over both first.

6.2.4 *Why the prosody field escapes it*

It is not rotated, and the module says why: its four columns are each a named thing, and rotating them would mix pitch back into the field the segment code was made shift invariant to get away from. Loudness, pitch against the speaker's own median, the movement of that pitch and voicing mean the same thing in every recording.

So the field that could carry the workbook's question is the comparable one, and the answer it gives is negative.

6.2.5 *A correction, recorded because the tree exists to prevent it*

This was first written and run as two speakers against one, on the reading that ICSNL58's filename names Givens and Hall where the other two name Bev Phillips. That was a guess off a filename, and the paper says who spoke: the story was recounted in Ntē?kepmxcín by Bev Phillips, who also provided much of the translation, and Givens and Hall are the linguists who transcribed and glossed it.

Guessing a speaker off a filename is the exact thing the rest of this tree refuses to do. `sift_extract.py` states the reason in its own header: a paper names its speaker in prose nobody has read, and that whole directory exists to avoid putting a name on somebody's words on no evidence. The rule was written down, and it was not followed here until the paper was opened.

The correction improved the result. Two speakers with one pair each would have given a comparison with a one in three ordering and no strength in it. One speaker across three recordings gives a necessary condition, and a necessary condition that fails settles the question.

6.3 THE WAVEFORM ITSELF, WITH NOTHING THROWN AWAY

Scope: `build/experiments/lossless_sound.py`, `build/sound/lossless/`.

6.3.1 *Why the field that already exists could not answer this*

`perceived_sound` reduces a recording to 28 bits every 10 ms and discards the rest. It takes an envelope, smooths the harmonic comb off it, keeps 24 coefficients, rotates them onto that recording's own singular vectors, and cuts every column at its median. Each of those is a bound, and the workbook's own entry on bounds is that every one put on a quantity in this work has had to be taken back off, ten times in one sitting, because the number then describes the choice.

The decoded waveform has no such choice in it. All three recordings are written out here at full integer width, in order, 215 MB of samples, and every reading below is taken off that. The bytes were checked against the corpus's signed audit first, on the standing warning that a measurement moving for no reason should be blamed on the bytes before the method.

6.3.2 *The first reading was wrong, and the samples say why*

Reading the top w bits of each sample returned $H_2(2)$ equal to $H_2(1)$ and $H_2(16)$ at 5.04 bits, far under what a speech waveform should carry.

The recordings say why. *When Old One Created the Earth* peaks at 16954 of 32768 and its median absolute sample is 6. Half of a twenty-two minute narrative sits inside thirteen levels of a sixty-five thousand level converter, because a told story is mostly the quiet between words. The top bits are sign extension: at two bits they take two values with the commonest holding 0.549 of the samples.

A linear slice of a waveform spends its levels where the signal is not. That is the same defect the workbook records four separate ways, and `envelope.py` already states the correction.

Two seatings are therefore reported. Linear over the range each recording uses, whose H_2 curve is a fact about the amplitude distribution. And quantile, which puts an equal share of samples in every symbol. H_2 comes out at w by construction. The second is the point: the marginal has been flattened to maximum entropy and every state is reachable. What is left to read afterward is arrangement and not loudness. It is the same subtraction radix makes on two dialects.

6.3.3 *Where the arrangement sits*

The block shuffle keeps everything shorter than its span and destroys everything longer. Sweeping the span locates the structure instead of only detecting it. The workbook builds this null on text and records it as never run on a waveform, and as unreachable with the earlier clips because measuring the arrangement of units needs recordings of tens of minutes. These are 3.9, 13.9 and 22.5.

Read on the quantile seating at eight bits, live against its own shuffle:

Span, seconds	<i>Moon</i>	<i>Sorry</i>	<i>Old One</i>	What that span is
0.00002	422.6	363.5	683.9	inside a glottal period
0.0002	48.0	46.9	71.8	inside a glottal period
0.0015	19.9	22.0	27.9	a pitch period
0.011	7.5	6.9	9.9	a segment
0.09	2.20	2.04	2.69	a phone
0.74	1.18	1.14	1.42	a word
5.9	2.83	1.05	1.07	a phrase

A ratio above one is the live recording carrying gaps more variable than its own shuffle, which is clustering. The workbook writes the same quantity as a ratio below one and the two are reciprocals. The *Moon* row at 5.9 seconds sits on 43 blocks and is scatter.

Three recordings, one curve, monotone across four orders of magnitude in span.

6.3.4 *There is no unit scale in it, and that absence is the result*

The obvious reading of that table is a knee somewhere near the word, and the text of one of these recordings is on disk to check it against: 1736 words over 1350 seconds is 0.778 seconds a word, arrived at with no reference to the audio.

Swept finely between a phone and six seconds, there is no knee. The excess over the shuffle decays smoothly and fits a power law in the span:

$$\text{excess} \propto \text{span}^{-0.752}, \quad r^2 = 0.988$$

over a 64-fold range of spans, with every one of eleven points landing between 0.88 and 1.19 of the fit. Nothing happens at 0.778 seconds. Nothing happens anywhere.

So the waveform's arrangement has no characteristic scale. That is a finding. It is not a failure to find one, and it is the case the workbook already names: data carrying structure at every scale has no characteristic scale for a window, a band or a ceiling to sit at. Every such bound in this work had to be removed. What holds still for such data is how the quantity changes with scale, and here that is an exponent measured to $r^2 = 0.988$.

6.3.5 *What that settles about reading speech back*

Detecting that a recording is speech is available and is not the interesting half. The departure from the null runs 363 to 684 at sample scale and is still 1.07 at six seconds. Nothing this large needs a careful instrument. Most of it is the amplitude envelope, which is to say the silence between words, and a measure reading that is reading that a person paused.

Transcription is not available from this reading, and the reason is structural. Writing down what was said requires knowing where one unit ends and the next begins. Section 4.12 of the method finds a boundary in text, by a symbol whose gaps are far tighter than geometric, and the whole of that construction rests on there being a scale where the regularity sits. This waveform has no such scale. A power law with no knee supplies no unit boundary at any span. There is nothing to segment on and nothing to write down.

That is the same blocker `examples/sound/README.md` already records from the other side, that the null has to keep each call intact while permuting the order of calls, and nobody has written it. This measurement says why nobody has: that null requires fixing the unit, and the unit is the thing being asked for. The block shuffle is the closest available substitute and it returns no unit.

Dialect cannot be asked at all. The audio held here is one speaker, one language, one dialect: Bev Phillips, `nle?kepmxcín`, Lytton. There is no second dialect to discern from the first, and a summed reference over one dialect is that dialect. The dialect question has an answer in this work and it is on the text side, where Mellesmoen and Kye's published labels supplied a border the algorithm never saw. Asking it of sound needs recordings carrying a border somebody else labeled, and the corpus holds none.

6.4 THE JOIN BETWEEN THE WEB AND THE SOUND

Scope: `build/experiments/align_oracle.py`, `build/experiments/pooled_sound.py`.

6.4.1 *The rung that was missing*

The refs chapter states the three stages and says which one is absent: the word web is built on written forms, and nothing takes a character out of `TEXT_SPACE` and hands back what it stands for. Two dialects writing one word two ways are still two nodes.

The section above went looking for the unit in the waveform alone and found a power law with no knee. That is the middle of the three ways a partition is fixed, an estimate, and the workbook says an estimate reaches nothing past what the sample already carries. The third way adds information the sample does not hold, and it is the only one that does.

6.4.2 *The supervision was printed on the page and copied into the table*

Hall and Phillips write that there is no standardized punctuation convention for nle?kepmxcín, instead of periods they separate each sentence of the transcription with a timestamp into the recording. The hand extraction copied those into the gloss column.

That is **126 sentences of Bev Phillips's own spíləxm, each carrying the interval of audio it was spoken in and the exact letters she said**, and nothing in this tree produced any of it. It has been sitting in the gloss column of a table that was checked against its paper in both directions.

Checked before it was used: the timestamps run forward, the last is 1336 s against a 1350 s recording, and the 126 sentences cover all of it. A sentence runs 2 to 40 seconds, median 10.

6.4.3 *A segment is a grapheme cluster and not a character*

w is not a segment, it is labialization on the consonant before it, and 'is glottalization on the base it sits over. Pooling per character would measure two halves of one segment as though they were independent.

Joining every mark and modifier to the base in front of it, the rule the oracles already use, turns 5552 characters over 35 distinct into **4839 segments over 50 distinct**. Twenty of the fifty carry a mark or a modifier: č k^w ḳ ḳ^w ṃ ṇ p̣ q^w q̣ q̣^w ẉ x^w x̣ x̣^w ỵ ẓ ẓ̌ é ɣ^w ɣ̣^w. The inventory grows under clustering, because k^w and k stop being one letter counted twice.

é is one of the fifty, at 45 instances. It is the same segment that carried the Lushootseed border, and at the character level it was two codepoints. The radix found it at width two.

6.4.4 *Placing a segment inside its sentence, and why the crude answer is allowed*

A timestamp bounds a sentence and not a segment. The segments inside one are placed evenly, which is wrong in detail, since a vowel is longer than a stop and nothing here knows that.

The comparison it is held against allows it. The null permutes which letter is attached to which interval and leaves the rest unchanged. Every interval keeps its position, its length and its audio, and the even spacing sits in both arms exactly. An error in the placement costs both arms the same and cannot manufacture a result. It also bounds the finding: what reads as holding together is holding together at the resolution even spacing allows.

6.4.5 *What the pooled sound gives*

Each interval becomes the loudness heard in 64 log spaced bands, out of the engine's own perceived_sound, normalized so that a loud instance and a quiet one of one segment are one

shape. A segment type's sound is the mean over its instances, and that is their sum. Thirty-four of the fifty types carry at least 24 instances.

a type's instances sit from their own mean	0.2766
two types' means sit apart	0.0591
between over within	0.214
the same letters shuffled over the same intervals, median	0.191
shuffles reaching the real letters	9 of 200

Two things have to be read off that together, and reading either alone gets it wrong.

The alignment is real. Nine of two hundred shuffles reach what the letters a person wrote down give. The letters and the audio are not independent, and the only thing joining them is a timestamp printed in a paper.

The letters do not separate. Between over within is 0.214, which is to say a segment's own instances scatter about five times further than one segment's mean sits from another's. A representation that recovered a letter from its sound would have that the other way up. This one does not, at this resolution.

So the join exists and carries something, and it is nowhere near reading a letter off the sound. The honest form of the result is that a rung which did not exist now exists, has a number on it, and the number is small.

6.4.6 *What it needs next, which the measurement names*

Even placement is the binding limit and it is fixable without giving up the supervision. The letters would still come from the oracle and only the boundaries would come from the audio, and the null would still permute labels over whatever boundaries were found. The control does not weaken. A sentence of ten seconds holding thirty-eight segments currently gives each one 0.26 s where a stop is nearer 0.05. Most intervals straddle a boundary or land in silence. That is where the scatter is.

What must not be done is to let the audio choose the letters. The timestamps and the transcription are everything that makes this supervision and not an estimate, and an alignment tuned until the letters separate would be the algorithm scoring itself against a split it proposed, which the workbook records as the failure that passes forever.

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RUNNING THE EXTRACTION

7.1 RUNNING THE SCRIPTS

Purpose: Go from an empty checkout to every corpus, check and sift result in this directory, without owning the archive first. **Scope:** data/salishan/

Run from the repository root. Everything writes under build/.

7.1.1 *Start here*

```
python -m pip install pypdf pypdfium2 requests numpy soundfile
python maint/data/salishan/get_papers.py
```

That fetches the eleven papers the readers need and writes build/papers/<name>.txt for each. Every other step in this directory waits on it, and it is the only step that touches the network. pypdfium2 renders a page to an image, which two of the eleven papers need and the other nine do not. numpy and soundfile are for the sound representation and nothing in the text pipeline imports them; soundfile decodes the mp3s in build/audio/, which are not fetched by this script.

--all fetches all 993 papers in the archive instead of the eleven. --list prints what would be fetched and stops. --convert converts PDFs already sitting in build/papers/ and fetches nothing, skipping any PDF that already has text beside it. --stem <name> takes one paper by its filename.

A converted paper whose fonts renumber their codes and declare no /ToUnicode map gets a .unfaithful file written beside its text, naming those fonts, and the run prints NOT THE PAGE on that line. The text is written anyway, because leaving a gap makes every later run try the paper again, and nothing downstream should read it as the paper. Six of the 146 PDFs on disk are in that class. theory/Salishan/chapters/chapter_Salishan_refs.tex names them and says what to do instead.

7.1.2 *Why there is a script for it*

The instructions used to say to download the PDFs, convert each to text, and name the text after the PDF. Two of those three steps have a wrong answer that looks like a right one.

The encoding. pdftotext writes Latin-1 unless told otherwise, and Latin-1 has no ?, no ə and no ı. A paper converted that way still opens and still looks like a paper, with the language taken out of it.

The reading order. Given `-enc UTF-8` the characters survive, and the page comes out in layout order, and a running header prints above the title it sits under and a two-column table interleaves. `pypdf` reads a page in the order the PDF stores it. The readers were written against that order, and `get_papers.py` uses it. Converting `Mellesmoen_Kye_ICSNL61.pdf` gives 1462 lines identical, in order, to the copy in `build/papers/`.

The script identifies itself by name, purpose and address in its user agent. The archive answers a request carrying none with an HTML page instead of the file; its `robots.txt` disallows only `wp-admin`, `wp-login`, the cache and trackbacks.

7.1.3 *Order*

Nothing after the fetch needs the network. The sift depends on the readers, because its second anchor is the corpus they produce.

1. `get_papers.py` `build/papers/*.txt`
2. `pdf2png.py` `build/pages/<stem>/page_NNN.png`
3. `draft_page_text.py` `build/papers/<stem>.page.txt`
4. the eleven readers `build/corpora/*.txt, *.pure.txt, *.unclassifiable.tsv`
5. `oracle_check.py` the hand extractions against the papers
6. `reader_check.py` the readers against the hand extractions
7. `coverage_check.py` reads 4, reports missing tokens
8. `english_sift.py` reads 4 for both anchors
9. `sift_extract.py` reads 4, writes `build/corpora/sifted/`

Steps 2 and 3 apply only to a paper the fetch flagged NOT THE PAGE, which is two of the eleven. For the other nine the extracted text matches the page and both steps are skipped.

7.1.4 *Where the tools are*

<code>maint/data/salishan/get_papers.py</code>	the archive
<code>maint/data/salishan/pdf2png.py</code>	the page as an image
<code>maint/data/salishan/draft_page_text.py</code>	the page as a first draft
<code>maint/data/salishan/hand_extraction/</code>	the control
<code>src/python/engine/nbody/orior/instrument/</code>	the algorithm
<code>maint/data/salishan/corpus_script_extraction/</code>	the eleven readers
<code>maint/data/salishan/orior_algorithmic_extraction/</code>	the sift applied
<code>maint/analysis/sound/</code>	the recordings as bits

Each script finds the repository by walking up to the tree holding `build/`.

7.1.5 *The twelve readers*

One per paper. Each takes no arguments and rebuilds that paper's corpus. Written `S/` for `maint/data/salishan/corpus_script_extraction/`.

```
python S/extract_garcia.py
python S/extract_hall_phillips.py
python S/extract_lafontaine_janzen.py
```

```
python S/extract_matthewson_redan.py
python S/extract_alexander_davis.py
python S/extract_mary_george.py
python S/extract_nater_bella_coola.py
python S/extract_lyon_priests.py
python S/extract_lindley_lyon.py
python S/extract_hilbert.py
python S/extract_mellesmoen_kye.py
python S/extract_robertson.py
```

Each prints its line counts, how many target-language forms reached the pure stream, and how many lines it could not sort. The two Lyon readers also report how many interlinear blocks read cleanly.

7.1.6 *The workflow the readers exist inside*

A reader is not the source of the corpus. The hand extraction is, and the reader is graded against it. Written H/ for maint/data/salishan/hand_extraction/.

```
python H/oracle_check.py
```

Checks each hand extraction against the paper it was read off, in both directions. A form written down that the paper does not hold is a typing slip. A word in the paper that no row holds is a row somebody skipped. A person reading a thirty-seven page paper into a table produces both. Both must be zero.

Two papers are checked against a drafted page text and the output line says so. For those, direction one measures the draft and not the table: a form the table holds and the draft does not is a place where a person read the page and a rule in `lyon_encoding.py` got it wrong. Direction two still means what it means everywhere else.

```
python H/reader_check.py
```

Checks each reader against the hand extraction. Grades whether it found each form, whether it says the same kind, and whether it says the same dialect. Kind decides the corpus. A rejected tableau candidate filed as a citation puts a form the paper's own analysis rejects into the pure stream, and nothing downstream asks again.

7.1.7 *When the text is not the page*

19-Lyon_ICSNL50_final-78 and 2013_Lindley_Lyon are set in a font that rennumbers its glyph codes and declares no map back to Unicode. Their `build/papers/*.txt` holds the font's alphabet and not what the page prints. `refs.md` has the detail, the other four papers in the same class, and why the damaged text is kept.

```
python maint/data/salishan/pdf2png.py 2013_Lindley_Lyon 1 70
python maint/data/salishan/draft_page_text.py 2013_Lindley_Lyon
```

The first renders each page to `build/pages/<stem>/page_NNN.png`. Arguments are the stem, the first page, the last page, and optionally a scale, which defaults to 3 and puts a 12pt body around 50 pixels tall, where a stacked diacritic stops being a guess.

The second writes `build/papers/<stem>.page.txt`, line for line with the extraction, which makes a page of one a page of the other. That file is a draft, and `oracle_check.py` reads it in place of the extraction for these two papers, which makes it the thing being graded and not the thing that grades. The rules are in `lyon_encoding.py`. Two of them cannot be settled without looking at the page: which `w` is a labialized consonant, and which inserted space is a word boundary. `refs.md` names those and the four other classes the check keeps reporting.

Both of these papers' readers take that file too, and so does `coverage_check.py`. They swap `page_text.py` in for `font_repair.py` to do it, because applying the mapping to text it has already been applied to destroys the text. Steps 2 and 3 therefore have to run before step 4 for these two, and running the readers without them leaves the last drafted text on disk in the corpus.

7.1.8 Checks

```
python S/coverage_check.py
```

Differs every language token in each paper against the corpus built from it. Both sides go through the same repairs first. Expected result is 0 missing on all eleven. For the two papers named above it reads the extracted text, where 0 means the reader got everything the font's encoding held, which is a weaker statement than it is for the other nine.

```
python S/font_substitution.py <damaged paper> <clean reference> [more]
python S/font_substitution.py 19-Lyon_ICSNL50_final-78 LyonICSNL60_Inch-2
```

Tests a candidate character mapping by counting how many damaged tokens become forms attested in a clean paper. The reading is the change between the two rates, and each rate alone is only half of it.

```
python S/case_delta.py
```

Matches the sifted output case-sensitively against the pure corpus. A form that matches only once case is folded is the same word written wrong, and the count of those is the formatting damage.

7.1.9 The sift

Written `A/` for `src/engine/python/instrument/` and `X/` for `maint/data/salishan/orior_algorithmic_extraction/`.

Note, 26 September. Commit 74601c1 deleted `src/engine/python/` on 26 September. At orior 1948ae1 there is no `A/` directory, and the commands below that run a script from it have no file to run. `corpus_derivation.py`, `X/boundary_check.py` and `binary_sound.py` are at the pin and import orior from `A/`. `perceived_sound.py`, in the table below, was deleted with the rest. Every result in this research paper was produced before the deletion. **Later, 26 September.** Commit d09b489 (orior PR 12) put back the 126 files 74601c1 deleted, each byte-identical to its state before the deletion, and biohub pins orior at d09b489. Every file named here that 74601c1 deleted, `A/` included, is at the pin again; nothing here was rerun against them.

```
python A/english_sift.py --check
```

Calibrates against the control. Prints how the known-pure corpus and the known-English spans separate, under one anchor and under two.

```
python A/english_sift.py <paper stem> [more stems]
python A/english_sift.py 1983_Hilbert
```

Scores one paper's lines against English and prints the twenty English accounts for least. A stem is a filename in build/papers/ without its extension.

```
python X/paper_language.py
python X/sift_extract.py
python X/language_check.py
python X/corpus_growth.py
python X/corpus_admit.py
```

The first reads which language each paper is about out of its own title and abstract. The second sorts every paper without a reader against both anchors and writes build/corpora/sifted/. The third asks whether the distributions agree with what each paper says it is about, with section 3 deciding whether the reading may be made at all. The last two print each corpus's split-half distance, support and entropy as it grows, and admit candidates in batches while the corpus stays on that curve.

7.1.10 *The sound representation*

Located at utils/maint/analysis/sound/binary_sound.py.

```
python utils/maint/analysis/sound/binary_sound.py
python utils/maint/analysis/sound/binary_sound.py <stem>
```

Writes build/sound/<stem>.bits.tsv in this tree, one row per 10 ms frame: the time, a 24 bit segment field, and a 4 bit prosody field. With no argument it does every recording in build/audio; a stem is a filename there without its extension. That repository's bytes hold still between runs. A signature over an inventory that moves on every run says nothing.

What it prints is the delta from a flat distribution over each field, at 6, 8, 10, 12, 14 and 24 bits, with how many states each width reached. Read the narrow widths. A 24 bit field has 16.8 million states and the longest recording has 135 thousand frames. The wide figure is about the frame count. refs.md has the measured numbers and the four facts about hearing the method is built on.

7.1.11 *What the modules are for*

These modules are libraries: other code imports them.

Module	Holds
A/orior.py	the method itself: squash, total variation, split-half, support, entropy the English anchor and the per-line screen built on it the T and N marking convention, tagged_spans, TEXT_SPACE, CAPS_RUN, ligatures what a reader needs where its source is the drafted page text: every repair a pass-through, and the language test asked on the orthography the flag file, and what counts as a language token the Lyon substitution table and the three forms of applying it putting back together words the Lyon extraction split internally closing the space a PDF leaves after a stacked diacritic the same, with a wider mark set that paper's two spaces allow
A/english_sift.py	
S/salish_marking.py	
S/page_text.py	
S/salish_unsorted.py	
S/font_repair.py	
S/space_repair.py	
S/inserted_space.py	
S/mellesmoen_kye_repair.py	

S/mary_george_repair.py
 S/glyph_names.py
 S/line_breaks.py

lyon_encoding.py
 H/papers.py
 src/engine/python/representation/sound/perceived_sound.py

the same again, where the grave is glottalization and the acute is stress
 the glyph names Robertson's extraction printed, read back as the characters they name
 putting back a word that extraction broke across two lines, shared by that reader and
 the coverage check
 the NimbusRomNo9L codes read as the orthography, for the drafted page text
 which hand extraction goes with which paper, its repair, and its marks
 a recording as bits: the log spaced bands, the audiogram, the shift invariant segment
 field, the prosody field, and the rotation that brings the sample to maximum entropy

TEXT_SPACE is the only definition of what characters these orthographies are written with. salish_unsorted.py and H/papers.py each held their own copy of that union until the copies were noticed; both read it from salish_marking.py now, and a per-paper set is written as that plus what the paper adds.

Author: dstroyo (Douglas Quigg) <dquigg123@gmail.com> **Date:** 2026-09-04

SOURCES

8.1 SALISH SOURCES

Purpose: Find the source behind any line in the extracted corpora, and know what is held on disk against what is only cited. **Scope:** build/papers/, build/corpora/, build/audio/, and data/salishan/

Everything here is generable using the data/ scripts or reachable at a stated address. Where a recording is known to exist but is not a member of the corpus, it is noted.

8.1.1 *Speakers*

These languages belong to the people who speak them, and none of this work exists without them. The corpus everything else here is measured against is their words, written down.

The list of who is in it is pure_corpus/README.md, which opens every entry with the speaker and carries the conditions they set. That file is written from maint/data/salishan/corpus_script_extraction/paper_config.py by maint/data/salishan/hand_extraction/pure_corpus_index.py. A speaker's name is therefore typed in exactly one place, and this document keeps no second copy of it to drift.

Fifteen of the twenty papers name their speakers. The other five cite a published dictionary and never say who spoke, and their entries say so. A linguist wrote the paper and a person read the paper into a table, and neither of those is whose language it is.

8.1.2 *Text sources, extracted*

Eleven papers, each read by its own file in data/salishan/corpus_script_extraction/ and each with a hand extraction beside it in hand_extraction/. Every token of the language in each is accounted for in the corpus built from it, and coverage_check.py says so. For the two Lyon papers that statement is about the extracted text, the font's own encoding and not what the page prints. The section below on those two says what that means.

A twelfth, 2012_Robertson, now has both. extract_robertson.py reads it, coverage_check.py puts it at 100 percent like the other eleven, and both hand-extraction checks grade it.

A thirteenth, 1975_Hilbert_Hess, is read by hand and has no reader. papers.py carries it so oracle_check.py grades it, and reader_check.py lists it under the papers waiting for one. The

section on papers whose extracted text is not the page says why a reader for it would have nothing sound to read.

A nineteenth, Hall-et-al_-ICSNL_61-1, prints every form twice and only one of the two was said. Hall, Luntzlara, Mellesmoen and Reid are arguing about what the *n̄eʔkepmxcín* transitive suffixes are underlyingly. Each example carries a surface form in square brackets and the underlying form they propose for it in slashes: [*wécentis*] beside /*wéce-n-t-ey-es*/. The hand extraction is 494 rows and verifies in both directions with nothing left over.

That layout makes the delimiter the evidence, and `extract_hall_ctr.py` reads only the delimiter. A bracketed run is a word and reaches the pure file. A slashed run is the authors' analysis, holds morpheme boundaries and a null-morpheme sign, and does not. A star marks a form the analysis predicts and the language does not have, written **[kícetxʷ]* in one place and *[*kícexʷ]* in another. The reader tests for both shapes. The delimiters stay on the form in the record, because the record holds what the page printed, and come off on the way to the pure file, where the entry is the word.

Two things the reader deliberately does not find. Examples (24) and (29) to (31) are ordered-rule derivations and every line between the underlying form and the surface form is a stage: *kícntene*, *kícntne*, *kícnne*, *kícne*. Those carry no delimiter, because the paper sets them in a column. The prose also prints some surface forms bare, *kícnəxʷ* and *wíktxʷ* among them. The hand extraction has all of them and the reader leaves them unclassified, which for a form nobody said is the safe direction to be wrong in.

Section 3.1.1 argues from two other languages, and the who column carries them: *ʔayʔajuθəm* has *-θi* and *St'át'imcets* has *-ci*, and Newman's proto-Salish **c* and **ci* are cited beside them. The reader puts zero forms under the wrong language. One string is genuinely ambiguous and is worth naming: *-ci* is both the *St'át'imcets* suffix the paper cites and the *n̄eʔkepmxcín* suffix the paper proposes. Nothing in the token separates them. *-ci* enters as *n̄eʔkepmxcín*, which the paper argues it is.

A bug this paper's coverage check caught in the reader. The reader files whatever its two patterns did not reach, and it built that leftover by taking the matched forms out of the line by name. Page 13 carries /ʔ/, which yields the one-character form ʔ, and taking that out of the page by name took it out of {*n*, *h*, *ʔ*}morpheme-t-___ as well. The spans come out by position now. One token still does not come back, and it is the same brace artifact the Lyon inchoatives paper has: ʔ]}morpheme-t-___ holds a }, the record delimits its spans with braces, and the check splits it on the way back out. The token is in the paper and in the record, and the paper stands at 99.7 percent for it.

An eighteenth, 2013_Nater, is the largest table in the set. Nater is asking how Salish Bella Coola is, and his answer is his whole lexicon sorted by where each word came from: 209 proto-Salish, 94 Coastal, 63 Interior, 214 non-Salish, 34 areal, and 661 of unknown origin. That is 1407 numbered entries over 52 pages, more of one language than the rest of this set holds together. The hand extraction is 2605 rows and verifies in both directions with nothing left over.

The fourth column of its two tables is whichever language Nater is comparing against, and `extract_nater_etymology.py` finds where that column starts by the token that opens it: one of the sixteen abbreviations section 1.1 defines, a language the body defines, or a reconstruction opening with an asterisk. Haisla, Heiltsuk, Oowekyala and Kwakwaka are North Wakashan. Proto-Athabaskan, Eyak, Carrier and Tahltan are Na-Dene. Nootka, Quileute, Chinook and Yurok are

four families past that. Squamish, Sechelt, Shuswap, Lillooet and Halkomelem are Salish and still not this language. The who column carries all of it, the reader puts zero forms under the wrong language, and only Bella Coola reaches the pure file.

The paper prints that column's own language at the head of every table page, as Line Gloss BC CS Cognates, and the reader reads it from there (`maint/data/salishan/corpus_script_extraction/extract_nater_etymology.py:80`). A bare `*x"əp` under that head is Coast Salish and holds nothing in itself that an explicit tag would have supplied. Reading the head moved 326 reconstructions out of the unclassified file and under the name of the group they reconstruct.

Section 4 prints each of its entries twice. `AKW'A` and `?ak"wa` are one word, and section 2.1 gives the mapping: `w` is written `w`, `c` is `ts`, `ʔ` is `tʃ`, `ʔ` is `lh`, `x` is `c`, `χ` is `x`, and `ʔ` is `7`. Both are Bella Coola and the table records both. Only the phonemic column reaches the pure file, because two orthographies in one stream would put the byte pairs of a transliteration into the measurement of a language.

One paper can write one sound two ways. The tables set every ejective as `"`, U+201F, 1763 times. Section 3's prose sets it as `'`, U+2019, 51 times. The prose's `t'nχw` and the appendix's `t'nχw` are one word in two encodings. The schwa does the same, `ə` U+01DD 183 times against `ə` U+0259 52 times, and NFC unifies neither pair. This is the same class of finding as the three characters for the lateral fricative and the two for the glottal stop below. This paper is the first here to hold both sides of that class, and a mark set built from the tables alone reads section 3 as English.

Its practical orthography also writes the glottal stop as the digit 7, which the shared mark set holds. A citation like `Sq67:137` or `Ku02:86-87` therefore carries a character of the language, and direction two asks about it. Three rows of the table exist for that reason. Nothing has been changed to stop it, because a digit that is a letter in one orthography cannot also be punctuation in a check that covers that orthography.

Its reader is graded at 2560 forms asked for, 703 not found, 2189 invented. Both of the large numbers are one structural difference counted from two sides: the table records each cognate as its own row, and the reader writes the cognate column as the single string the paper printed. `*x"əp` „deep (water)“ is one item against two rows. That is the Lyon papers' disagreement inverted, and it is the shape the section on hand extraction describes.

A seventeenth, `Kim_TwanaReduplication_final`, is the first Twana paper here, and Twana had no anchor at all before it. Its hand extraction is 362 rows and its reader writes 252 Twana forms. Every Twana form in it is Drachman's, out of a 1969 dissertation the paper calls the only reliable reference in existence for Twana CVC reduplication. What is on these pages is close to everything written down.

Kim also carries the trap this set makes easiest to fall into. Example (1) sits where the data usually starts and it is Tillamook, in Edel's 1939 transcription with its own vowel letters; (19) and (20) are Thompson and Lillooet; footnote 9 mixes Puget Sound Salish, Moses-Columbian and a row of English phonetics. The who column carries all of it and the footnote is left unclassified, because the language changes inside one sentence there.

The lateral fricative has a third character. `salish_marking.py` says papers differ between `ʔ` and `ʔ` and carries both. Kim writes it `ʔ`, U+026B, sixty-two times. A mark set holding the other two reads every one of those tokens as English. The same paper writes the glottal stop two ways, `ʔ` where it is phonemic and `ʔ` where a rule derived it, which its footnote 11 states outright.

A sixteenth, `LyonICSNL60_Inch-2`, is Lyon's survey of how 48 Nsyilxcn roots inchoativize, elicited from `łkmxnalqs` Delphine Derrickson-Armstrong and `čəskʔáknaʔ` Dave Michele of `stqəʔtkʷłniwt`. Its hand extraction is 537 rows and verifies in both directions. Most cells of its two tables are starred, which is a form the linguist built and the speakers rejected; those are held out of the pure stream, because a smaller corpus is better than one seeded with words the language does not have. The reader writes 295 Nsyilxcn forms and files the Spokane, Secwepemctsin, Lillooet, `nxaʔamxčín`, Thompson and Coeur d'Alene cognates under their own languages.

This was the first paper here below 100 percent coverage, at 99.5, and the two tokens are worth naming because neither is a word. `closed_spaces` closes the space after a glottalization mark, and on page 9 that mark ends a real word twice, welding `√puł` and `púł` onto the `qt: {` that opens the dictionary quotation after them. The welded tokens hold a brace, the record delimits its spans with braces, and `coverage_check` therefore splits them on the way back out. They exist in neither the page nor any corpus. The corpus is correct not to hold them, and the check is correct to say it does not.

A fifteenth, `ICSNL59_Nater_2_final`, is Nater arguing that Bella Coola has words with no vowel in them, and its data is 127 numbered entries that are each a run of two to four consonants. Six more entries and two comparison tables are Heiltsuk, Oowekyala, Kwakwala and Haisla, which are North Wakashan and not Salish at all. The who column is doing more work here than anywhere else in the set. Its hand extraction is 671 rows and verifies in both directions with nothing left over.

A fourteenth, `WolfeICSNL60`, is a different shape from all of them and has both. It is a comparative reconstruction. Instead of one speaker telling one story it cites lexical suffixes from published dictionaries of eighteen languages, laid out in columns under a two-letter abbreviation. Its hand extraction is 772 rows and verifies in both directions with nothing left over. The section below on what a pure corpus is for says why it writes a `.pure.tsv` keyed by language where every other reader writes a flat `.pure.txt`.

Nothing here needs fetching by hand. `python maint/data/salishan/get_papers.py` reads the archive index, downloads these eleven and converts them. Every address below was checked; the local filename in `build/papers/` is the PDF's own name.

Paper	Reader	PDF
Three Glossed Nleʔkepmxcín Narratives by Kʷəłtəzətkʷu (Bernice Garcia). Garcia, Hannon and Stacey. ICSNL 59, 2024 [4]	<code>extract_garcia.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2024/07/ICSNL59_Garcia_Hannon_Stacey_final.pdf
ł cutəs us ł qəłmin ł tmiɣʷ (When Old One Created the Earth). Hall and Phillips. ICSNL 60, 2025 [7]	<code>extract_hall_phillips.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2025/07/HallPhillipsICSNL60.pdf
Four Stories by wɬwlmelst. LaFontaine and Janzen. ICSNL 59, 2024 [13]	<code>extract_lafontaine_janzen.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2024/07/ICSNL59_LaFontaine_Janzen_final.pdf
Cw7aoz káti7 láti7 ku naxwít (There was definitely no snake there). Matthewson and Redan. ICSNL 61 [19]	<code>extract_matthewson_redan.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2026/07/Matthewson_Redan_ICSNL61.pdf
ł Tsiɣwas sQwa7yán'ak Áku7 Graveyard Valley. Alexander and Davis. ICSNL 61 [2]	<code>extract_alexander_davis.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2026/07/AlexanderDavis_ICSNL61.pdf
Mary George Personal Narratives. John Hamilton Davis. ICSNL 56, 2021 [11]	<code>extract_mary_george.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2021/08/ICSNL56_DavisJ_2_final-1.pdf
A Bella Coola tale: The Frog Children. Nater. ICSNL 50, 2015 [22]	<code>extract_nater_bella-coola.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2018/01/22-Nater-Bella-Coola-tale-10.pdf
Three Okanagan stories about priests. Lyon. ICSNL 50, 2015 [17]	<code>extract_lyon_priests.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2018/01/19-Lyon_ICSNL50_final-78.pdf
12 more Upper Nicola Okanagan narratives. Lindley and Lyon. 2013 [14]	<code>extract_lindley_lyon.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2018/01/2013_Lindley_Lyon.pdf
Poking Fun in Lushootseed. Vi taqʷsəblu Hilbert. ICSNL 1983 [9]	<code>extract_hilbert.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2018/03/1983_Hilbert.pdf
A Comparative Analysis of Stress in Northern and Southern Lushootseed. Mellesmoen and Kye. ICSNL 61 [20]	<code>extract_mellesmoen_kye.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2026/07/Mellesmoen_Kye_ICSNL61.pdf
Lexical Suffixes and Connectives in Proto-Central Salish and Beyond. Wolfe. ICSNL 60, 2025 [25]	<code>extract_wolfe.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2025/07/WolfeICSNL60.pdf
Voiceless Words in Bella Coola: Fact vs. Fiction. Nater. ICSNL 59, 2024 [23]	<code>extract_nater_voiceless.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2024/07/ICSNL59_Nater_2_final.pdf
Nsyilxcn Inchoatives and their Distributions Across Root Types. Lyon. ICSNL 60, 2025 [18]	<code>extract_lyon_inchoatives.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2025/07/LyonICSNL60_Inch-2.pdf

The truncated reduplication in Twana. Kim. ICSNL 52, 2017 [12]	<code>extract_kim_twana.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2017/07/Kim_TwanaReduplication_final.pdf
How Salish is Bella Coola? Nater. 2013 [21]	<code>extract_nater_etymology.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2018/01/2013_Nater.pdf
Ctrl-Alt-Delete: The Control Directive and Associated /t/ Deletion in nle?kepmxcin. Hall, Luntzlara, Mellesmoen and Reid. ICSNL 61 [8]	<code>extract_hall_ctr.py</code>	https://lingpapers.sites.olt.ubc.ca/files/2026/07/Hall-et-al.-ICSNL_61-1.pdf

Each reader writes into `build/corpora/`: the marked record, a `.pure.txt` holding only target-language speech, a `.unclassifiable.tsv` listing what it could not type, and for the two Lyon papers a `.words.txt` giving the word forms recovered from the interlinear.

Records are named speaker first: `<spoken by>_<paper>_<who wrote it down>_Salish_<language>_<year>_<mixed>`. The speaker is who the corpus is of.

8.1.3 Hand extraction

The reader is not the source of the corpus. Each paper is read by a person into `maint/data/salishan/hand_extraction/<paper>.oracle.tsv`, which says for every form in the paper what it is, and the reader is graded against that.

Sixteen of the twenty verify against their paper in both directions: no form written that the paper does not hold, no token in the paper that no row covers. The other four are the papers whose extracted text is not what the page says, and what their remaining disagreements measure is the damage in that text. The two sections after this one are about them.

Kim was the fifth until this pass. Its four direction-one failures were the extraction doubling a combining mark on a phonetic form, and closing them by transcribing the doubled form would have put a corrupt string in the table. `paper_config.KIM_DOUBLED` corrects the four at the source instead, which made that paper faithful and moved it into the sound count.

One hole in direction two is not a hole in a table, and Nater’s paper is where that shows. His argument is that Bella Coola has words with no vowel in them. Entries like `kp` ‘each, all, every’, `tp` ‘spotted’, `ps`, `px` and `sx` are plain ASCII with nothing in them that any mark set can catch. `is_language_token` needs a marked character, and about thirty of the 133 entries are invisible to direction two. Direction one still checks every one of them, because it looks for what the table wrote down. The apostrophe was tried as a way in, since Nater writes every ejective with one; it also makes every English gloss a word of the language, because a gloss ends in a closing quote and the two characters are the same. That trade is a few hundred false holes for about thirty real ones. The apostrophe stays out and the limit is recorded here instead.

Counts below are what `oracle_check.py` and `reader_check.py` report on 2026-09-04. “Asked for” is how many distinct written forms the rows yield, the denominator for the column beside it. “Invented” is a form the reader put in the corpus that no row asks for.

Paper	Rows read by hand	Forms asked for	Reader
Mellesmoen and Kye, ICSNL 61	467	443	reproduces it exactly
Hilbert, ICSNL 1983	60	52	2 not found, 8 over-runs it flags itself, 1 invented
Matthewson and Redan, ICSNL 61	104	103	44 not found, 217 invented
Alexander and Davis, ICSNL 61	541	532	339 not found, 914 invented
Nater, ICSNL 50	285	285	29 not found, 80 invented
LaFontaine and Janzen, ICSNL 59	226	225	81 not found, 136 invented
Garcia, Hannon and Stacey, ICSNL 59	371	369	150 not found, 748 invented
Mary George, ICSNL 56	1661	1654	663 not found, 574 invented, 362 in the wrong dialect
Hall and Phillips, ICSNL 60	406	399	209 not found, 521 invented
Lyon, ICSNL 50	1417	1413	1037 not found, 2134 invented, 193 in the wrong dialect
Lindley and Lyon, 2013	653	653	312 not found, 996 invented, 198 typed differently

Robertson, 2012 [24]	195	193	92 not found, of which 59 are prose the reader does not read
Hilbert and Hess, 1975 [10]	74	74	no reader written yet
Wolfe, ICSNL 60	772	678	153 not found, of which most is front matter and section prose the reader does not read; 224 invented, 0 in the wrong language
Nater, ICSNL 59	671	288	136 not found, the cluster charts the reader does not read; 337 invented, 0 in the wrong language
Lyon, ICSNL 60 (inchoatives)	537	491	76 not found, 61 invented, 0 in the wrong language; 146 typed differently, the table's own column vocabulary against the reader's one kind
Kim, ICSNL 52	358	295	67 not found, 124 invented, 0 in the wrong language; 18 the reader flagged instead of guessing at
Nater, 2013 (etymologies)	2605	2560	703 not found, 2189 invented, 0 in the wrong language; 280 typed differently. The two large numbers are one difference counted twice, named in the section on this paper above
Hall, Luntzlara, Mellesmoen and Reid, ICSNL 61	494	468	169 not found, 117 invented, 0 in the wrong language; 72 typed differently. What it does not find is the front matter, the prose-cited affixes, the derivation columns and the rules, none of which carries a delimiter
Davis and Mellesmoen, ICSNL 58 [3]	441	441	no reader written yet

Only Mellesmoen and Kye reproduces its table.

Wolfe's wrong-language column carries the most weight, because it is the only paper here holding more than two languages. That column reads zero. The who column carries the language on every row and the reader keys its output to the same column, and a Sechelt suffix cannot arrive in the Twana data by falling through a branch.

2013_Lindley_Lyon is read off the rendered pages: all twelve texts, both the running transcription and the interlinear, with the footnotes, free translations, commentary, the appendix paradigms and the abbreviation list. What remains is 46 forms the table holds that the draft does not, and 37 strings in the draft that no row holds. All 83 fall in one of the five classes the section below names, which makes the count a measurement of the draft.

Its reader takes `build/papers/2013_Lindley_Lyon.page.txt` and is graded in the orthography the table is written in. The reader misses 312 of the 653 rows, and 56 of those 312 are a single token. The other 256 are a row holding a whole translation, parse line or narrative paragraph, and the reader writes each of those as separate items instead of as the one string the row asks for.

19-Lyon_ICSNL50_final-78 is read the same way and is finished. The 1417 rows cover the title, the abstract, all three stories as running text and again as interlinear, the twenty-five footnotes, the section metadata and the references. What remains is 41 forms the table holds that the draft does not, and 32 strings in the draft that no row holds, and the two lists are the same finding twice: every one of the 32 is a form the draft got wrong and the table therefore does not carry.

Its reader misses 1037 of the 1413 forms, and 5 of those are a single token. The rest are a gloss line, a parse line, a word gloss or a running paragraph held whole, against a reader that writes the interlinear a token at a time.

Three of the eleven carry a `.oracle.md` beside the table, which is where a note about that paper goes. The `.tsv` itself holds no comment syntax and no prose header, which lets a CSV linter read it: a header on line 1 and the same field count on every row.

`oracle_check.py` tests the hand extraction against the paper both ways. `reader_check.py` tests the reader against the hand extraction.

`corpus_derivation.py` turns what those two and `coverage_check.py` report into a bound on the per-line error rate, and rewrites `corpus-derivation.md` from it after every change to a table. That script reads the checks by running them and parsing their own output, and never recomputes what they compute, because an earlier version reimplemented the same quantities and drifted: it reported 16 direction-one failures where `oracle_check` reports 0. The bound now stands at $2.73\text{e-}11$ per line across three channels. One of the three has seen a failure and is reported at its observed rate; the other two are at the rule-of-three bound, over 14255 and 7290 trials. The bound has moved the wrong way twice: from $6.03\text{e-}11$ up to $6.83\text{e-}11$ when a reader turned out to be losing forms, and from $3.06\text{e-}11$ up to $4.01\text{e-}11$ when the Hall paper arrived carrying a token its record cannot give back. Both rises are the measurement working. A bound that only ever falls is measuring the papers that were easy to add. The bound fell to $2.73\text{e-}11$ when the Kim corrections closed the last four direction-one failures and moved that paper into the sound count. The first significant figure has now settled, and the last three papers agree on it. The document's section 4 names the three places its independence is thin. The reader counts are deliberately not one of the three channels: a reader is written for one paper. Its error rate is a fact about that paper, and pooling eighteen of them would report a rate nothing is sampling.

Two errors it found that coverage could not, because coverage was 100 percent through both:

- The Hilbert record credited Vi Hilbert as the speaker. She wrote the paper; the twenty-one examples were said by her aunt **Susie Sampson Peter** of the Upper Skagit and by **Martha LaMont**, recorded by Leon Metcalf between 1950 and 1958 and by Thom Hess in 1963.
- Example 2's English translation is a single line, "High class, high class was Raven." The record had it with the next eleven lines of Hilbert's commentary welded on, as something Susie Sampson Peter said. The typescript sets examples in an indented column and the essay at full width; the extraction dropped the indent and the width recovers it.
- **Six of the eleven PDFs break words in half.** They leave a space after a stacked diacritic. *kweswapáw* arrives as *kweswapáw*. Counts: Hall and Phillips 996, Garcia 943, LaFontaine and Janzen 478, Mellesmoen and Kye 298, Matthewson and Redan 169, Mary George 159. The coverage check cannot see it, because it puts both sides through the same repair and a word broken on both sides matches itself. `inserted_space.py` closes it; " is held out, because it is a spacing letter and a space after one is a real boundary.

The stress accents are held out of that repair. A word can end in a stressed vowel, and *neʔé e sqyéytn* in LaFontaine and Janzen closes to *neʔée*, which the language does not have. Mellesmoen and Kye passes its own wider set instead, because that PDF prints two spaces at a real boundary and a lone space after an acute there is always the inserted one.

Matthewson and Redan carries a second form of the same damage that no rule can close safely. The PDF also breaks *before* a marked letter: *cáclep* arrives as *các lep*. The same shape is a real word boundary at *lta qíłqa*, and nothing in the characters separates the two. The reader leaves both alone; the hand extraction has the true forms. That paper's reader is also blind to sections 1.2 and 1.3, where the story's title and five *St'át'imcets* forms are cited in prose.

Three more the hand extraction found, none of them visible to a coverage check:

- **reader_check was reading a fraction of each record.** Its span pattern rejected any kind holding a space, and nine of the eleven readers write one: symbol note, running speech, stage direction, word gloss, free translation, cited example, morpheme entry, speaker comment, orthography chart.
- **The apostrophe is a letter.** ' is glottalization in Nuxalk and in Lyon's Okanagan, and stripping it as punctuation turned the enclitic `_c'` into `_c`.
- **Nater's reader kept section 2 as six blocks of prose** where the paper lists about forty prepositions, articles, deictics and enclitics, each with its own gloss, and never read section 1, where *sma* and *sʔalac'i* are defined.

8.1.4 Papers whose extracted text is not the page

Six of the 146 PDFs in `build/papers/` carry a font that renumbers its glyph codes with an `/Encoding/Differences` array and declares no `/ToUnicode` map. An extractor is then handed code numbers with nothing to turn them into, reads them in a default encoding, and what can land in the `.txt` is the font's private alphabet.

Every one of the six has now been opened against its own rendered pages, and they came out four different ways.

Paper	Fonts declaring no map back to Unicode	What the page says happened
19-Lyon_ICSNL50_final-78	NimbusRomNo9L Regu, Medi, Regultal	damaged, and read off the pages in full
2013_Lindley_Lyon	NimbusRomNo9L Regu, Medi, Regultal	damaged the same way, and read off the pages in full
Lyon-final [16]	NimbusRomNo9L Regu, Medi, Medital, Regultal	damaged, by a different table
2012_Robertson	Symbol and five TrueType subsets	damaged four ways, one of them decodable exactly
21-Abraham_ICSNL50_final-4 [1]	NimbusRomNo9L Regu, Medi	clean
2011_Lonsdale_Matsushita [15]	Courier, Times-Roman, Times-Italic, CMSY10	clean

A missing `/ToUnicode` is not the fault by itself. 141 of the 146 hold a font without one and nearly all of them extract correctly, because a standard encoding already says what the codes mean and every extractor has that table. It is the renumbering that does the damage, and 6 of those 141 renumber.

The renumbering is a risk. Two of the six extract correctly, and both were checked line for line against a rendered page.

21-Abraham_ICSNL50_final-4 is St'át'imc in the van Eijk orthography, which is ASCII apart from the accented vowels. Its renumbered codes never reach a character they would damage. All four pages come through, down to the apostrophes: page 1's first line reads *Ats'xenlhkán ta sásqets áku7 Nséq'a* (Charlie Mack's), *lti Líl'wata Tsel'álh c.walh*, on the page and in the text.

2011_Lonsdale_Matsushita is clean for a different reason: it prints its Lushootseed in the ASCII transliteration its own parser reads. There is nothing on the page for a font to lose. Page 5 sets *LEFT-WALL ?u+ da?a +d ?ElgWE? ?E kWi s+ gWistalb ti?E? SukWE? .* and the extraction gives that string exactly. The `?` is the glottal stop, *E* the schwa and *W* the labialization, on the page as much as in the file.

The list says which files to open. The page says what happened to each.

Page 23 of 2013_Lindley_Lyon prints

cítx^wsəlx uł fi nyʔip ckaʔítət

and `build/papers/2013_Lindley_Lyon.txt` holds

cítxws@lx uì 'ti ny 'Qip c 'kaPít@t

Every schwa is @, every lateral fricative ì, every pharyngeal Q, every glottal stop P; the wedge stands in front of its letter as ˘x, the ejective mark stands in front of its letter instead of over it, and the word ckaʔítət is split. pypdf and pypdfium2 lose the same things, which makes this a property of the file and not of one library.

One part of the loss cannot be inverted. Labialization is written with a raised w and the page also has a plain w. Both arrive as w. Page kʷukʷ and page wist are the same string in the text, and nothing in the file separates them.

The consequence for method is why this section exists. A hand extraction taken from one of these .txt files records the font's encoding and not the paper, and it then verifies clean against that same file in both directions, because both sides of the check are reading the one damaged artifact. That is how this was found here, after two such tables had been built.

What replaces it: pdf2png.py renders the pages to build/pages/<stem>/page_NNN.png and the reading is done from the image. draft_page_text.py writes build/papers/<stem>.page.txt, a draft of the page in the shared orthography, line for line with the extraction, which makes a page of one a page of the other, and lyon_encoding.py holds the rules it applies. The draft is a starting point a person corrects against the rendered page, never a source. Two things it cannot decide:

- **Labialization**, for the reason above. The drafter writes ʷ after the consonants that take one, which is right for kʷ qʷ xʷ ʃʷ and wrong wherever a real w follows a consonant. The rule errs in both directions, and page 315 has one of each. púlxiwi gets a ʷ it should not have, and section 3.2 settles that by parsing the word √púlxi-wi. lkʷu t loses the one it should have, because the inserted space split it into lk and wu t before the rule could see the consonant, and section 3.2 parses that one √lkʷ=ut. The rule runs over the whole line and reaches the English as well: stanza 117's word gloss he.fell.off.backwards comes out he.fell.off.backʷards.
- **Word boundaries.** The PDF inserts a space in front of a letter carrying a mark. s 'plá 'ks@lx is one word and iP 'kl is two, and both are a space in front of a marked letter. Page 25 settles the first as s'pláks@lx, page 24 the second as iP 'kl.
- **The space that was not inserted.** Where the PDF leaves the mark against its letter there is nothing to key on, and a medial ' is also the apostrophe of Lyon's. nkʷ'ritkw, xw'tilx and wa'y keep the mark standing in front of its letter; the pages read them nkʷ'itkw, xʷ'tilx and wa'y.
- **A P-initial word with no other mark on it.** Pitx, Pums, Pistkm and Paksyilx come through as English, because a capital P opening an otherwise unmarked token is the shape of Papers and Penticton.
- **The dropped space.** At a change of font and at some line ends the PDF loses the space entirely. Okanagan yaxʷt, uł ny, məł ixíʔ and axáʔ ikłcítxʷ arrive welded into one token.
- **A run of the length mark before a closing quote.** The raised dot arrives as ;, and a rule turns it back wherever a letter or a spacing mark follows it, because a lone ; is also Lyon's semicolon. A run of eight is never punctuation, and staʔʒíləm "whoa " is the only

place where a run is followed by a closing quote instead. That pattern occurs twice, both in 19-Lyon_ICSNL50_final-78, and 2013_Lindley_Lyon holds no run at all.

- **A one-letter gloss label that is also a glyph code.** A gloss token is recognized by a run of two capitals, because one capital on its own is also iP and Philosophical (lyon_encoding.py:116). Stanza 108's gloss line labels the question marker Q, one capital. The token falls through to the Salish branch and comes out ɿ. Nothing inside the token separates the two readings, and the drafter does not track which line a token sits on. That label occurs twice, both in 19-Lyon_ICSNL50_final-78.

All 83 disagreements the check reports on 2013_Lindley_Lyon fall in these classes now that it has been read off the pages in full. The last two classes occur in the priests paper alone. That count is the measurement the draft exists to produce.

One thing the extraction is better evidence for than the rendered page. The transposition moves a mark off its letter and preserves it, and a token's mark count survives the channel exactly. At the scale these pages render, a run like ṁṁṁá cannot be told from ṁṁá by eye, and reading the page alone undercounts. The text settles it: stanza 112 arrives as s- ṁ ṁ- ṁ ṁ-á ṁ- s, which is five marks. The word is s-ṁṁ-ṁṁ-áṁ- s. Take the count from the text and the boundaries from the page.

get_papers.py applies the same test inside converted() and writes a .unfaithful file beside the text naming the fonts. The test fires only when a PDF is converted, and conversion skips any PDF that already has text beside it. The table above was produced by running the same test over them directly.

A scanned typescript is a worse case than any font

1975_Hilbert_Hess carries no font problem at all. It is a 1975 typescript, scanned, and what sits in build/papers/ is OCR of the scan. Page 1 prints

k^wa? ləcuṁṁcadadid dx^w?al ti?ə? s?acuss (h)əlg^wə? dəx^wṁṁbil ?ə ti?ə?.

and the text holds

kWa? lacuXacadadid dxW?al ti?a? s?acuss (h)algWa? daxwxabil ?a ti?a?

Every schwa is a, every raised ^w is W, the wedge over x is gone, and sud^zig^wicuts comes through as sudZigWi0ltS. The six pages give 200 forms the table holds that the text does not.

The difference from the Lyon papers matters more than the size of it. There the damage is a font's renumbering, which is deterministic: one code always stands for one glyph, which lets a table invert it, and lyon_encoding.py is that table. Here the damage is a reader guessing at shapes on a scan, and sud^zig^wicuts to sudZigWi0ltS is not a substitution anything can run backwards. There is nothing to build. The hand extraction is not a check on the extraction for this paper, it is the only correct record of it.

That also puts a floor under the whole method. A paper this old cannot be verified in both directions, because one direction has no sound source to verify against. What oracle_check.py reports on it is the distance between a page and its OCR, which is worth having and is not the same measurement it makes elsewhere.

One defect the paper exposed in the check itself. oracle_check.bare() strips ? as punctuation, and this orthography and the 1983 typescript's both write the glottal stop as ?. So ?ə bares to ə

and a word with a glottal stop cannot be told from one without. This defect is the same class as the apostrophe case the hand extraction already found. The answer has to be per paper, since ? is punctuation in every other paper here. Nothing has been changed.

Page 6 of that paper is a comparative appendix in Upper Chehalis and Cowlitz, thirteen sentences of two other Salishan languages. The table names each of them in its own column, which keeps a reader from putting them in a Lushootseed corpus.

The delta is a mutation probability distribution

Everything above treats the damage as a hazard to route around. The damage is also the one calibrated mutation sample in the archive, and the damaged text is kept for that reason.

Two encodings of one text, aligned line for line, with the page settling which is right, is a measured channel. `build/papers/<stem>.txt` is the mutated side, `.page.txt` is the drafted side, and the hand extraction read off the rendered page is ground truth for both. No other paper here has a second encoding of the same page to compare against.

The mutations sort into five kinds, and the last two are what make this a distribution instead of a rewrite table.

Kind	Instances	Invertible
Substitution, one symbol for one	$a \rightarrow @, i \rightarrow 1, \zeta \rightarrow 0, ? \rightarrow P, \lambda \rightarrow \tilde{n}, \cdot \rightarrow ;$	yes
Collapse, two symbols onto one	$w \rightarrow w$, and plain $w \rightarrow w$	no
Transposition	a combining mark over its letter becomes a spacing mark before it: $k \rightarrow 'k, \tilde{x} \rightarrow 'x$	yes
Insertion	a space appears in front of a letter carrying a mark	not decidable alone
Deletion	the space at a change of font is dropped, welding two words	not decidable alone

The insertion does not always fire. `2013_Lindley_Lyon.txt` holds `wa 'y` in most places and `wa'y` in three, for the single word the page prints as `wa'y`. So its rate lies strictly between 0 and 1, and the same input shape has two outputs. That is a probability, not a rule, and it is why a repair applied without the page welds words that were never one word.

The deletion fires where the paper changes font mid-sentence. Footnote 5 of the same paper sets its cited forms in italic inside roman prose, and the two arrive as `0kanaganyaxwt` and `0kanaganyaŷáŷt`, each one token where the page prints two words. The second is recoverable, because `yaŷáŷt` is a common word and appears on its own elsewhere in the paper. The first is not: `yaxwt` occurs nowhere else. Which of the two happens depends on the rest of the corpus and not on the token. That is the second reason this is a distribution.

The collapse sets a ceiling. No recovery, however good, separates page `kwukw` from page `wist` in the mutated text, because the channel destroyed the distinction instead of disguising it. Anything downstream that claims to have recovered a labialized consonant on these two papers is claiming something the file cannot support.

A detector keyed to a glyph inventory reads the mutated side as holding no language at all. That is not hypothetical: `is_language_token` did exactly that on these two papers until the marks set for them was corrected, and it reported a paper with eleven unread texts as fully covered. A detector reading the distribution of symbols survives the same input. The pure corpus says how well a vector separates the target language from the English around it. This pair says how much mutation it takes before it stops separating them at all.

Nothing in the tree measures the per-symbol rates yet. `font_substitution.py` scores a candidate mapping by how many mutated tokens become forms attested in a clean paper, which is a

proxy for the answer. The page-verified table is the answer itself for the lines it covers, and it grows one text at a time.

The distribution was going to convert the rest, and that has now been tested. Four of the six papers above are set in NimbusRomNo9L, and the two with page-verified tables beside them supplied the candidate mapping for the other two. Both of those were checked against their own rendered pages. Neither takes it, for two different reasons.

21-Abraham_ICSNL50_final-4 has nothing to convert, for the reason the section above gives. `font_substitution.py` on it is measuring a paper that is already right.

Lyon-final is damaged and wants a different table. The paper is John Lyon on the linguistic evidence for a Francis Drake landing in Oregon, and the language data in it comes from Frachtenberg's Oregon transcriptions. ì for ɬ carries over and so does the transposed acute, but the paper also needs a transposed macron the Okanagan papers never use, E for ε, and several more marks. One code disagrees outright. ; is the raised length dot in the Okanagan papers and the glottalization mark here. Page 24's k'eu'ts! arrives as k;eu'ts!, and applying the Okanagan table to this paper would silently turn every glottalization into a length mark. `font_substitution.py` puts 4 of 407 occurrences into attested forms against 5 before it, the same answer measured from the other side.

So the conversion table is per document and not per font family. Reading these two papers off the page still recovers two papers and still produces a table, and what that table converts is the two of them.

2012_Robertson is damaged four ways, and one of them costs nothing. Its extractor could not resolve a glyph and printed the glyph's name instead. The text holds /uni029400 /uni026C /xé/ uni0294 where page 30 sets a morphemic line. A /uniXXXX name carries the code point it stands for, and /uni0294 is ʔ, /uni026C is ɬ, /uni0259 is ə and /uni019B is λ. `glyph_names.py` turns all 103 of them back, along with /combiningdotbelow and /combiningacuteaccent, and after it runs no glyph name is left standing anywhere in the paper. That part is arithmetic and exact.

The other three came off the pages, and the first of them is why the decode is not the whole recovery.

- **Labialization, collapsed exactly as it is in the Lyon papers.** Page 30 sets /kʷú[·kʷ]piʔ and the text gives /kwú[·kw]pi/uni0294. The paper's own prose loses it too: page 30 names its symbols (č, š, xʷ) and the text holds (č, š, x w), with a space where the raised letter was. Every flag the hand extraction raises on this paper so far is this one.
- **A word broken across two lines with no hyphen.** 42 lines of the extraction hold nothing but a fragment: Da above vid D. Robertson, ma above ximal efficiency. Most of them are in the interlinear, where the break falls inside a gloss: A above UG-, s above ick, gr above eet. A rule that joins every fragment line to the one under it would be wrong in the alphabet tables, where wa, wi and waw are Chinuk pipa letter names and each is a line of its own.
- **A glyph repeated where the page prints it once.** Page 1's epigraph prints t'íxʷəɬ γ-ʔ-[χ-]q'y-n'-tén with one ɬ and one ʔ, and the text holds four of each. The doubling happens on 2 lines in the paper and both are set in bold italic.

The hand extraction is 195 rows and covers both Salish texts whole, the Chinuk pipa and Chinook Jargon forms cited through the analysis, and the two bibliography entries written in Chinuk

Wawa. What it leaves is 40 forms the table holds that the text does not, and 21 strings in the text that no row holds. Those two lists are the same finding twice, as they are in the priests paper: 14 of the 21 are the damaged form of something the table carries correctly, 4 are the doubled epigraph, and 3 are page ranges in the bibliography that `is_language_token` reads as `language` because they hold a digit inside a token.

Texts 3 through 6 are Chinook Jargon, which is a pidgin. They are the paper's own material and they are not Salish. The table marks who each form belongs to for that reason, and Chinook Jargon in that column is the instruction to a reader to keep the form out of the target stream.

`extract_robertson.py` reads it against that table. The reader finds all 20 stanzas of Text 1 and all 7 of Text 2 with their numbering intact, and it reproduces the four rows of a stanza wherever the extraction did not break a word inside a line. Of the 92 rows it does not reproduce, 59 are the analysis prose and the cited forms, which the reader does not read and which stay in the record as unclassified. The 33 that sit inside the two texts split as 10 differing from the reader by the lost labialization alone and 23 by a word the extraction broke in the middle of a line, where the first half ends a long line and the reader has nothing short to key on.

Two things in that reader are worth naming because no other paper here needs them. The transliteration row is plain ASCII. `o l ha l kukpi`, carries no character of the modern orthography and the test every other reader leans on would call it English. The row is marked by where it sits. And the pure stream takes the transliteration row alone: the morphemic row is Robertson's analysis and holds forms nobody wrote, the distinction `salish_marking.py` draws between what was said and what was worked out afterward.

`line_breaks.py` holds the join, because `coverage_check.py` has to apply it too. That check compares a paper against its extraction and its own header says both sides go through the same transformation first, and a join the reader makes and the check does not reports every welded word as a hole. With it applied the paper is at 100 percent like the other eleven.

`lyon_encoding.py` holds that table for `NimbusRomNo9L` as it stands, with the labialization rule still a guess and the insertion rate still unmeasured.

The algorithm reads the damage the blanket repair cannot

`maint/data/salishan/orior_algorithmic_extraction/symbol_sift.py`. Reading a broken word off a rendered page works and does not scale: twenty papers at twenty-four pages each is four hundred images, and every reading is a judgment nobody can check afterward. This asks the sift instead.

A break site is a place where the extraction has two tokens and the paper may have had one. The candidates are the two joined by each combining mark the paper writes, and joined by nothing, the reading where the space is a real boundary. A candidate counts as attested where the paper writes it as a token, and where the paper writes it inside one: `qʷaɫ út` is `qʷaɫút`, which Davis and Mellesmoen never prints alone and does print inside `qʷəqʷaɫút`. Scoring is the radix from `boundary_check.py`, with run counts taken from the tokens carrying no break and flattened to maximum entropy. A run common everywhere contributes nothing and what is left belongs to that restoration.

It finds 86 sites across the fifteen papers with readers, and every one of those papers already reports 100 percent coverage. That is not a contradiction, it is the blind spot inserted_

py covers applied to it. The measurement that says so is 2013_Lindley_Lyon, whose table is read off the pages in full and which takes no repair. The table and the corpus compare token for token with nothing in between.

	reading the extraction	reading the page text
corpus tokens the table holds	1231 of 1878, 66%	2204 of 2535, 87%
marks standing in front of their letter	1832	54
labialized consonants written with a plain w	1630	42

The counts in the last two rows are for both Lyon papers together. The corpus used to hold *iks'ma'yítím* where the page holds *iksma'yítím*, and *sqwlqwlstwíxwtət* where the page holds *sqwlqwlstwíxwtət*. The corpus now holds what the page holds.

What is left of the 13 percent is the inserted space: a word arrives in pieces and the corpus keeps the pieces. *incítxw* is in it as *incítx w*. `space_repair.py` exists to close those and cannot reach these, because the vocabulary it joins with comes from the interlinear's form column, which in this paper is the parse and carries morpheme hyphens the running text does not.

The two readers swap `page_text.py` in for `font_repair.py`. Every repair in it returns its line unchanged, because applying the mapping to text that has already been through it destroys the text. P to ? turns Papers into ?apers and the gloss label APPL into A??L. Its `language_line` and `carries_orthography` had to be rewritten instead of passed through, because `font_repair`'s versions ask whether a line holds a character only the damaged orthography writes, and the page text holds none of them by construction.

`coverage_check.py` reads the same page text for these two, under a page entry in its repair list, and all twelve papers with readers are back to 100 percent coverage.

What a pure corpus is for

The purity of both sources is the constraint everything else answers to. The target is the Salishan ingestion and the reference corpus is English, and a boundary drawn between two distributions says nothing about the languages if either side is holding something other than what it claims.

A paper holding eighteen languages has no single pure stream, and Wolfe is the first one here that does. Every other reader writes a `.pure.txt` of one language, because every other paper is one speaker telling one story. Wolfe cites lexical suffixes from published dictionaries of Sliammon, Sechelt, Squamish, three kinds of Halkomelem, three kinds of Straits, Klallam, Lushootseed, Twana, Tillamook, Quinault and two Tsamosan languages. Pouring those into one file builds a corpus of no language at all, and it would be a corpus the sift is then measured against, where that mistake cannot be recovered from at all. So `extract_wolfe.py` writes a `.pure.tsv` with the language in the first column and no flat pure file, and `language_check.BY_CORPUS` has no entry for the paper. Nothing joins an anchor by accident.

Two further things are held out of that file. A reconstruction is a form the author worked out and nobody has ever said one, the distinction `salish_marking` already draws between a transcription and a segmentation line, applied to a proto-form. And the predicted reflexes of Tables 6 and 8, which give for each language the form expected if the reconstruction was stressed, the form expected if it was not, and the form actually attested; the extraction does not keep those three columns apart reliably. Nothing is lost by that, because the attested forms of both tables are printed again in examples (1) and (6) where the columns are unambiguous.

The oracles can also be turned on themselves. Every one of them is verified against its own paper. They should agree with each other about what these languages look like, and one of them can be measured against the rest of them taken together as the reference corpus. A row that sits oddly against everything else verified is one of two things: a slip in the transcription, or a real property of that paper nobody has noticed. Both are worth opening the paper for, and neither is visible from inside one table.

The leave-one-out shape of that now runs at the language level. `corpus_derivation.py` measures each dialect against the other five pooled into one reference corpus, the question the sift is actually asked, and all six stand off the pool. That shape does not yet run per row, and running it per row would name a paper to open.

Once each dialect's corpus is pure, the dialects become each other's reference corpora. A boundary drawn between Nsyilxcən and Ntəʔkepmxcín is a claim about where one ends and the other begins, and it is worth drawing because of what sits near it: forms that two dialects share and a third has lost, which is where onomatopoeia that has gone out of one of them would show.

One such boundary has now been drawn and scored against an answer the algorithm did not supply. Lushootseed is not one dialect, and Mellesmoen and Kye's stress paper labels every form it cites northern or southern. The hand extraction copied that label into the `who` column. The border sits on disk as a published fact, from one paper, in one orthography, by one transcriber. `maint/data/salishan/orior_algorithmic_extraction/boundary_check.py` loads those labels, sets them aside, and only compares at the end. Comparing whole distributions fails at this size and says so: the two varieties hold 1469 and 2768 bytes against a method that resolves at 6707, and a blind partition scores exactly the majority class. Holding the concept fixed with the web's gloss edge, flattening the pooled run counts to maximum entropy, and testing one run at a time does find it. The term is the stressed schwa é, southern, 45 against 6 over the 58 concepts both varieties name. Putting the border back at random 200 times, 1 of those random borders finds as much. `corpus-derivation.md` section 7 carries the tables.

That comparison has to run on a sound representation in binary, not on the characters. These orthographies write the same sound differently and the difference is arbitrary. The lateral fricative is ʔ in one paper here and ʕ in another. The glottal stop is ʔ in most and the digit 7 in the van Eijk papers. Labialization is a raised w in some and a plain w in others. A model counting character bigrams reads two dialects that share a word as maximally far apart, because it is measuring the transcriber and not the speaker. Encoding each segment as its features and comparing those makes the distance a fact about the languages.

The plan those two things sit inside runs in three stages. First the pure text corpus, which fixes what each dialect writes and which the whole hand extraction workflow above exists to produce. Then a sound representation measured off recordings, where a segment is a set of binary features and the alphabet in `TEXT_SPACE` maps onto those features one written symbol at a time. Then a probability distribution over that representation, the thing an interdialect boundary is drawn on. The word webs are built on the encoded forms. Two dialects writing one word two ways arrive at one node.

The recordings are what the sound representation is measured from, and `build/audio/` holds three of them. All three are Ntəʔkepmxcín and two are Bev Phillips. That is enough to build an encoding against and short of what a second dialect would take. The section below names the recordings that exist and are not published with their papers.

Stage two is built and the section after this one gives what it measures. The word web is built as well, on the written forms: `maint/data/salishan/word_web/word_web.py` joins every form in the hand extractions to the forms it is related to and writes one file per group under `build/corpora`. The web has three kinds of edge. A **concept** edge joins two forms whose glosses share a content word. It crosses an orthography, because the gloss is the only part of a form two papers wrote the same way. A **shape** edge joins two forms sharing a leading or trailing run of four characters, which in these languages is usually a shared root or affix. A **context** edge joins two forms written in the same section of one paper by one speaker. The web stands at 7062 forms and 373575 edges across nineteen groups, up from 3382 and 37484 before the Nater etymologies joined it. The edge count grew ninefold on twice the forms, and a dictionary does that to a web keyed on glosses: 1275 entries each carrying an English gloss give the concept edge far more to join than the same number of words in running narrative.

What has not been built is the join between the web and the sound representation: nothing takes a character out of `TEXT_SPACE` and hands back the features it stands for. The web's nodes are still written forms, and two dialects writing one word two ways are still two nodes. `TEXT_SPACE` in `salish_marking.py` is the inventory of characters these papers are written with, which is that join's other side, and it stays one definition instead of a copy per file for that reason.

`english_sift.py --check` reports the separation as healthy against that corpus. Under one anchor, 2285 known-pure lines sit at a median of 15.15 against 1899 known-English spans at 8.49, and a cut at 9.28 keeps 99.0 percent of the pure corpus while admitting 15.3 percent of the English. Under two anchors, 96 percent of the pure lines come back as language and 98 percent of the English spans as English. Those figures say the two anchors are far apart. They do not say the language anchor is the language, and `orior-salishan.md` has what the six per-language anchors do, where they fail to separate, and why.

The same measurement on the other nine papers is not comparable and should not be read as one. Each of those goes through its own repair before the comparison, and several of those orthographies write labialization with a plain `w`.

The binary sound representation

`src/engine/python/representation/sound/perceived_sound.py` turns a recording into two bit fields per 10 ms frame, and `binary_sound.py` writes them to `build/sound/<stem>.bits.tsv`. Four facts about hearing set its shape, and each of them is in the code.

The waveform on its own is not enough. A vowel is a harmonic series under an envelope, and two recordings of one vowel share the envelope while the harmonics sit wherever the speaker's pitch put them. The analysis is spectral and the envelope is smoothed off the source before anything is compared.

What a person hears is not what the microphone recorded, and it differs between listeners by physiology and by how much hearing they have lost. The weighting is an argument, with `NO_LOSS` standing for a listener who has lost none. A real audiogram is the six frequencies a hearing test is run at with the loss in dB at each, and the loss comes off a band before loudness, and a band a listener cannot hear stops reaching the code.

Some sounds are the same sound at different frequencies. A phone said by a small speaker and by a large one differs by a scaling of the whole spectrum. The 64 bands are log spaced, which

makes that scaling a shift along the band axis, and the magnitude of a Fourier transform along that axis is unchanged by the shift. That transform is the segment field.

And some sounds that differ only in frequency mean different things. Stress carries an acute in every orthography in this corpus, and a field made shift invariant to buy the paragraph above would put a stressed form and an unstressed one at one address. The prosody field is separate for that reason and holds loudness, pitch against the speaker’s own median, how that pitch is moving, and voicing.

Bringing the sample to maximum entropy makes the delta readable. Each bit is cut at its own column’s median over the recording and splits the frames in half, and the segment columns are rotated onto their principal axes first so that no bit restates another. All 24 segment bits come out set on 0.500 of frames on all three recordings, and all 64 of the six bit states and all 256 of the eight bit states are reached on all three. A recording that used those states evenly would be noise, and the distance from that flat reference is the structure in it.

Measured on 2026-09-04 over the three recordings in `build/audio/`, with the entropy and total variation out of `orior.py`:

Recording	Frames	8 bits	10 bits	12 bits	Prosody
Givens and Hall, ICSNL 58	23489	0.2157	0.2538	0.3161	0.3708
Hall and Phillips, ICSNL 59	83249	0.2364	0.2863	0.3207	0.3498
Hall and Phillips, ICSNL 60	134997	0.1859	0.2200	0.2519	0.4594

Each number is the total variation between the codes that width of field took and the flat distribution over its states. The width is part of the figure because the field has to be sampled to be read. At 12 bits the longest recording puts 33 frames in each of 4096 states; at 24 bits it puts 133820 codes in 16.8 million states, which measures the frame count and not the recording. The run reports 6, 8, 10, 12 and 14 bits for that reason, with how many states each width reached.

All three recordings came from the `nle?kepmxcin` Lab and two of them are Bev Phillips. These numbers are the method working and not a comparison of dialects. A second dialect’s recording turns them into a reading.

8.1.5 Audio sources

Recording	Address	Held
Hall and Phillips, xʷiʔ kʷ páq (You Will Be Sorry), ICSNL 59 [6]	https://lingpapers.sites.olt.ubc.ca/files/2024/07/ICSNL59_Hall_Phillips_audio.mp3	yes, 13.4 MB
Givens and Hall, The Moon and the Birchbark Canoe, ICSNL 58 [5]	https://lingpapers.sites.olt.ubc.ca/files/2023/07/ICSNL58_Givens_Hall_The_Moon_and_the_Birchbark_Canoe.mp3	yes, 7.5 MB
Hall and Phillips, ɬ cutés us ɬ qəlmín ɬ tmixʷ, ICSNL 60 [7]	https://lingpapers.sites.olt.ubc.ca/files/2025/07/2025-05-29_BP_storyrevised_trimmed.mp3	yes, 17.1 MB

The third is Bev Phillips reading the story `extract_hall_phillips.py` reads. It is an oracle for that extraction and not only a source.

Recordings named in papers and not published with them:

- Qwaʔyánʼak, Graveyard Valley narrative. Just over half an hour, recorded by Henry Davis at Nxwísten on 7 July 2025, published with the ICSNL 61 paper.
- Kʷeswapáw (Linda Redan). Three minutes twenty-eight seconds, told over Zoom on 31 October 2025. Audio and video are held by her.
- George Lezard, 1966, recorded by Randy Bouchard.

- Nellie Guitterez, 1978 or 1979, recorded by Yvonne Hébert at an Elders' Gathering in Quilchena.

8.1.6 *Text sources*

build/papers/ holds 152 extracted paper texts and 146 of the PDFs they came from. Eighteen have a reader and a hand extraction and one more has a hand extraction alone. That leaves 133 of the 152 unread and 841 of the archive's 993 not fetched. Six of the 146 are in the class the section above describes, and their texts are the font's encoding.

Fifty of them are about Lushootseed or Skagit. The heaviest, by how much they discuss it, are Mellesmoen_Kye_ICSNL61.txt, Beck_2007.txt, 1994_Beck.txt, 1995_Beck.txt, 1998_Cort.txt, 2000_Barthmaier.txt, 1996_Beck.txt, ICSNL58_Kye_final.txt, 01_ICSNL55_Beck_final.txt and 2002_Lonsdale.txt. 1983_Hilbert.txt is in that set.

Which paper to read next

paper_language.py reads each paper's front matter for the names the literature uses for each language and attributes a paper where it names one and no other. Over the 154 texts it attributes 122 and leaves 32, which are the comparative papers and are correctly left. Lushootseed leads at 43 papers, Nuxalk follows at 32.

Counting how much of a language each unread paper actually holds gives a different order, and it is the order to read by. Lushootseed has 40 unread papers and 593 distinct language tokens across all of them together, which is less than one Nater appendix. The reason is the section above: most of the Lushootseed literature is older and writes the glottal stop as ? and the schwa as a or E, and a mark set built on the modern orthography reads those papers as holding no language at all. Comox has 8 unread papers and 1899 tokens, St'át'imcets 10 and 1297, Nuxalk 29 and 1046.

So the count of papers per language is not the count of language per language, and picking by the first number would spend the next month on texts a check cannot yet see into. What Lushootseed needs is not another paper but the treatment 1983_Hilbert and 1975_Hilbert_Hess already have, which is a per-paper mark set for the orthography that paper is actually written in.

8.1.7 *Verification sources*

Sources used to test a transformation instead of to supply text. Each answers a question the code that made the change cannot answer about itself.

- LyonICSNL60_Inch-2.txt. Lyon's later paper on Nsyilxcən, whose extraction kept its characters. Used twice: to test the font substitution table, which moved attested tokens from 1 of 3599 to 811 on one paper and 2 of 4332 to 965 on the other, and to test the word list recovered from the interlinear, where none of the broken forms had its pieces attested apart and 4 percent and 36 percent were attested once joined.
- Each paper's own second printing. Hall and Phillips, Garcia, and both Lyon papers print their stories twice, once as running text and once word by word. One printing checks a reading of the other.

8.1.8 *Community sources*

- Puyallup Tribal Language Program, <https://www.puyalluptriballanguage.org/>
- The Salish Institute, <https://www.thesalishinstitute.com/salish-language>

8.1.9 *Where the volumes live*

- **Every paper, one page, 993 links:** <https://lingpapers.sites.olt.ubc.ca/icsnl-volumes/>. ICSNL 1967 to present. The 152 texts in build/papers/ came from here, and anyone without them starts here.
- UBCWPL, <https://lingpapers.sites.olt.ubc.ca/>. Volumes 33 (1998) to 61 (2026).
- ICSNL archives, <https://blogs.ubc.ca/icsnl/icsnl-archives-and-precedings/>. Volumes 1 to 32 are the Kinkade Collection.
- nłeʔkepmxcín Lab, <https://blogs.ubc.ca/nlab/projects-publications/>. Where the three audio addresses came from.

Whole volumes: ICSNL 50, <https://lingpapers.sites.olt.ubc.ca/files/2018/01/ICSNL2015-fullonline.pdf>.

Paper addresses follow files/<year>/<month>/<name>.pdf, and <name> is the local filename in build/papers/. UBC answers a command-line fetch with a bot-defense captcha; a browser gets the same file.

Compiled By: dstroyo (Douglas Quigg) <dquigg123@gmail.com> **Date:** 2026-09-04

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A reader who disagrees with a number in this research paper can ask which copy it was checked against, and the hash answers exactly. Two copies of a paper are rarely the same bytes.

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